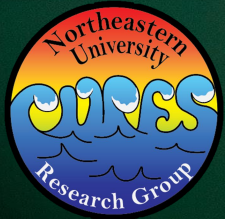


Relating Wave Geometry and Surface Dynamics to Subsurface Velocities

Tyler McCormack, mccormack.ty@northeastern.edu
Dr. Julia Hopkins, ju.hopkins@northeastern.edu



Presentation Preview

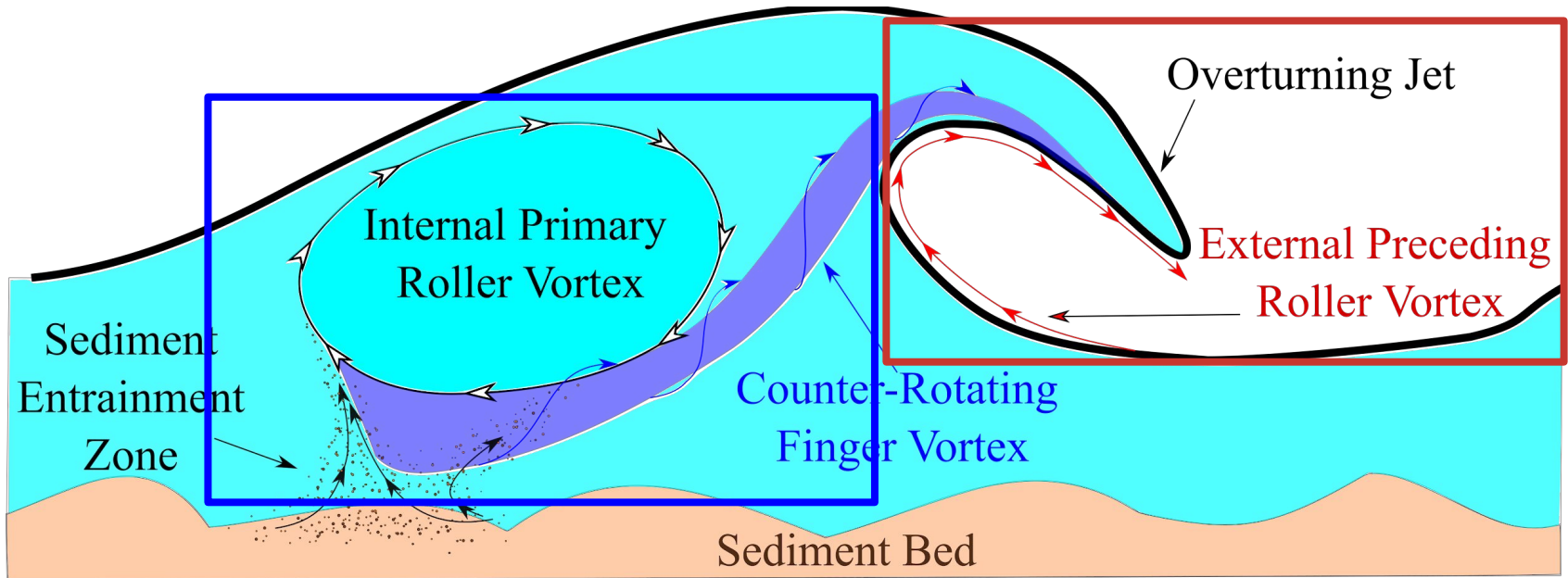
1. Research objective
2. Previous work
3. SPAM experimental setup and timeline
4. ADCP-LiDAR Analysis
5. Conclusion

Background

Goal: Predicting subsurface velocities with surface characteristics

Background

Goal: Predicting subsurface velocities with surface characteristics



Background

Using surface velocity to predict subsurface velocity

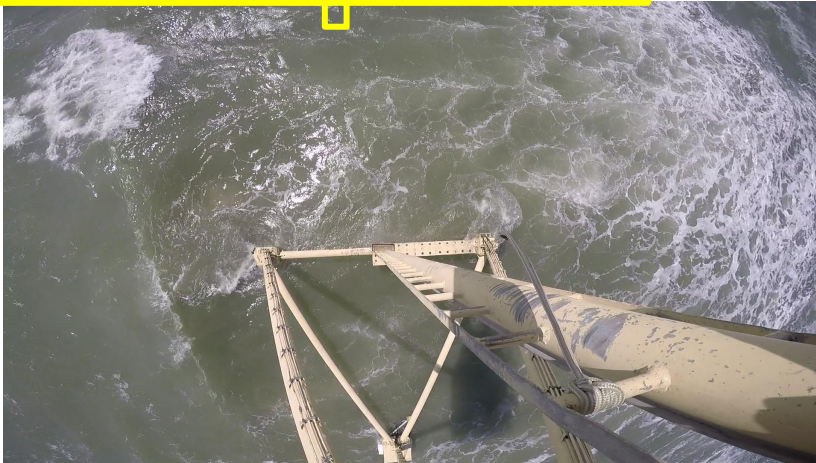
Background

Using surface velocity to predict subsurface velocity

Data Provided by:
Steve Elgar
Britt Raubenheimer
Levi Gorrell
Kate Brodie
FRF Ops Team



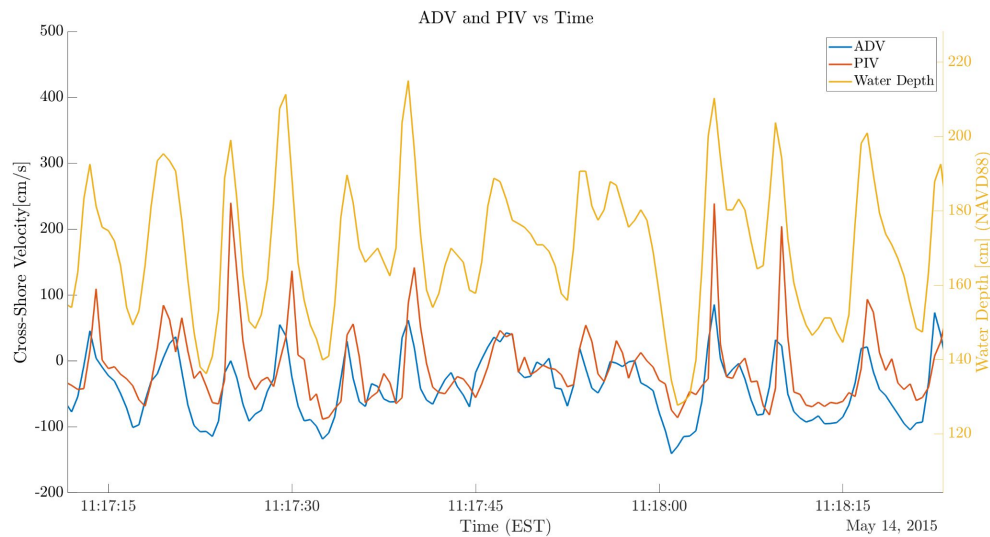
Particle Image Velocimetry (PIV)
measurement region



Background

Using surface velocity to predict subsurface velocity

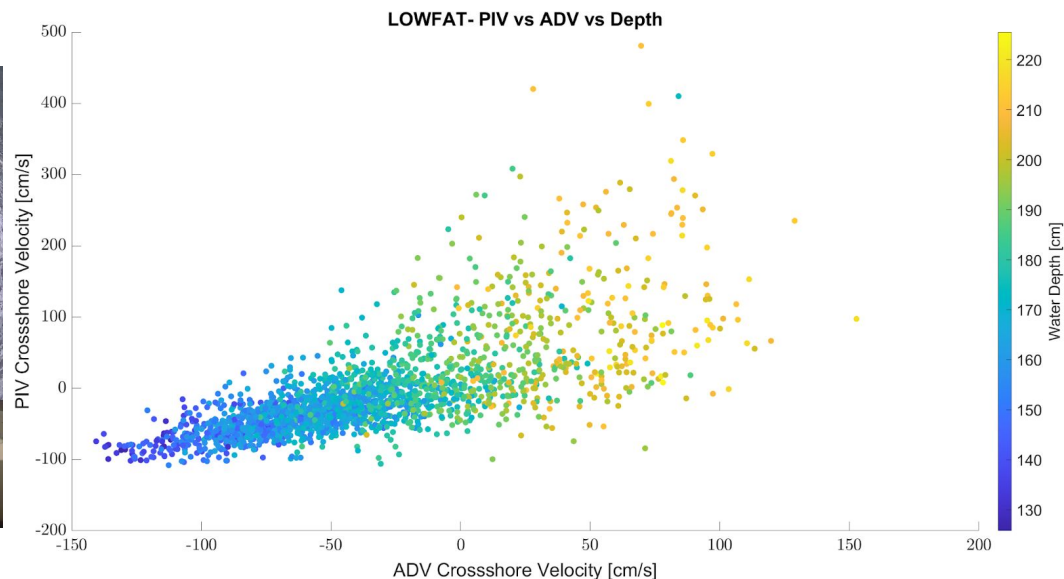
Particle Image Velocimetry (PIV)
measurement region



Background

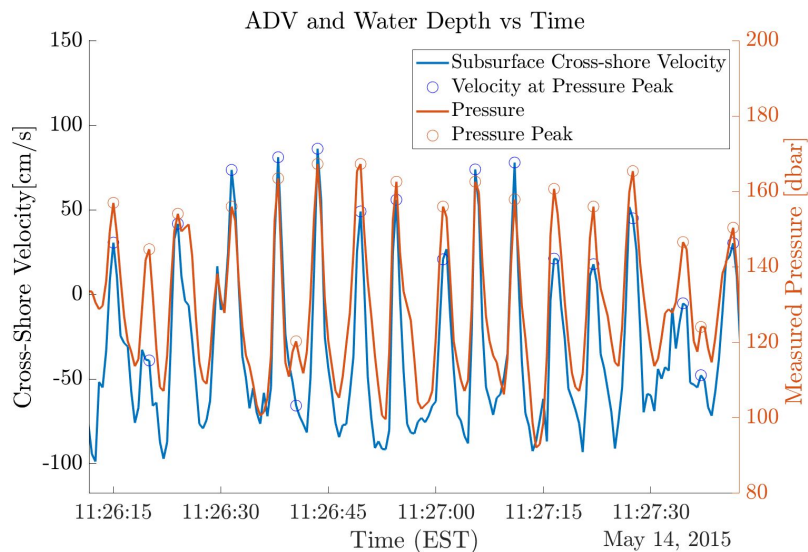
Using surface velocity to predict subsurface velocity

Particle Image Velocimetry (PIV)
measurement region



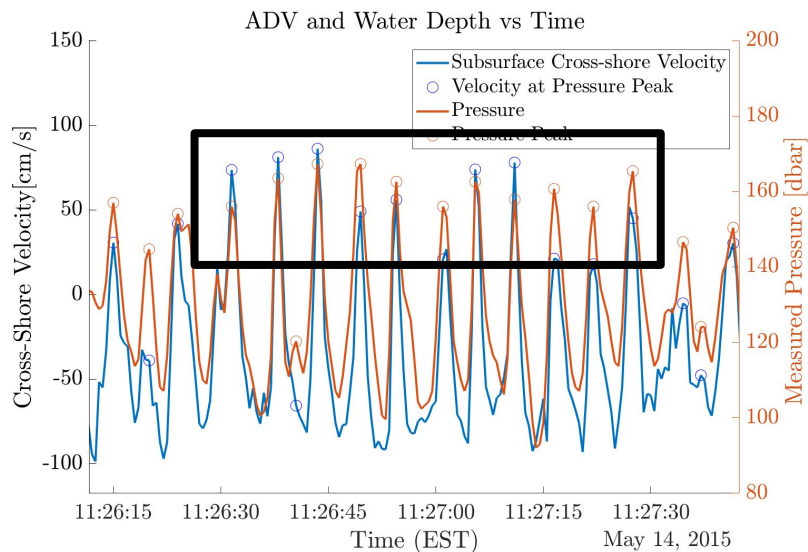
Background

Using pressure data as a proxy for wave height to predict subsurface velocity



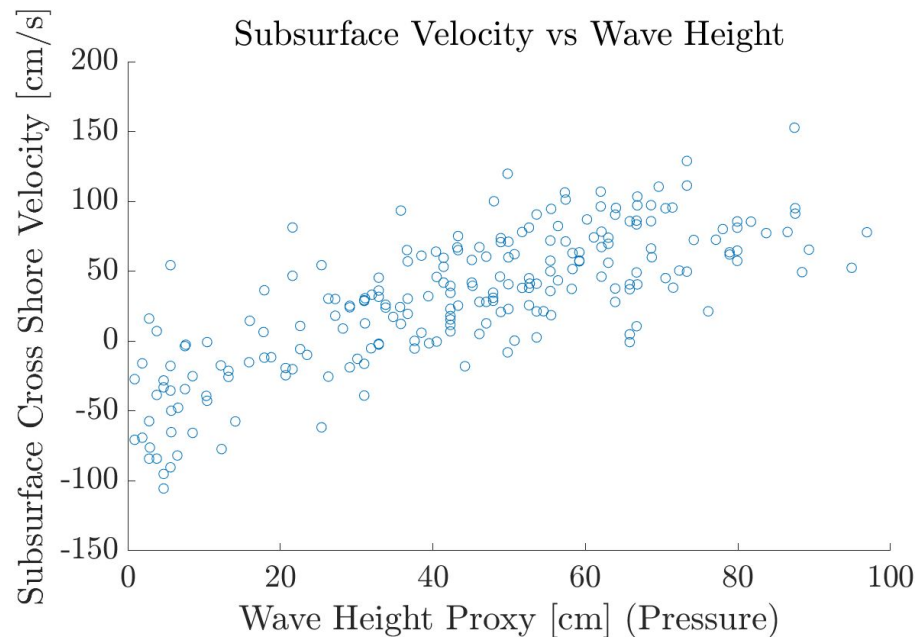
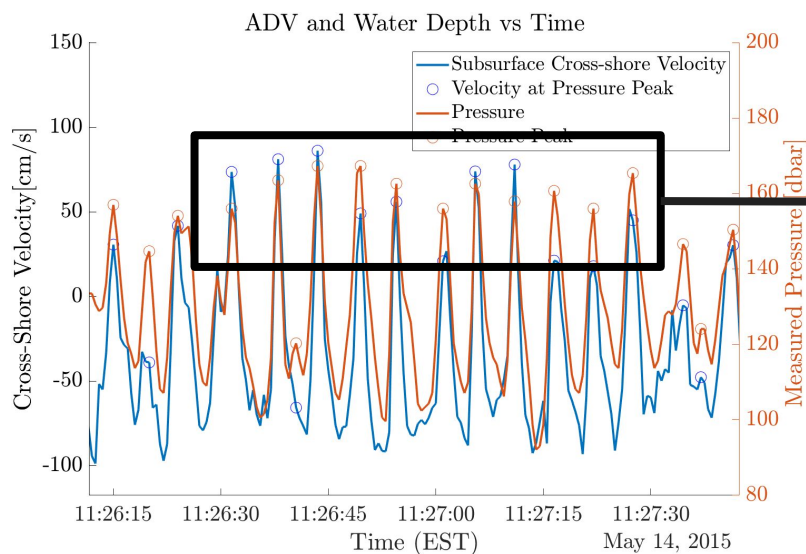
Background

Using pressure data as a proxy for wave height to predict subsurface velocity



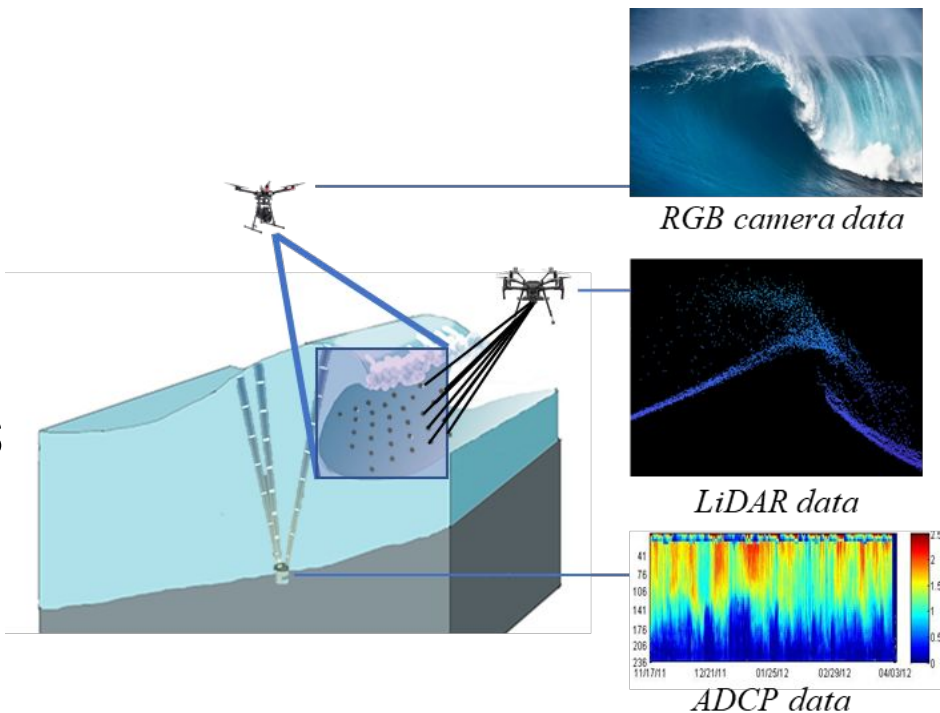
Background

Using pressure data as a proxy for wave height to predict subsurface velocity

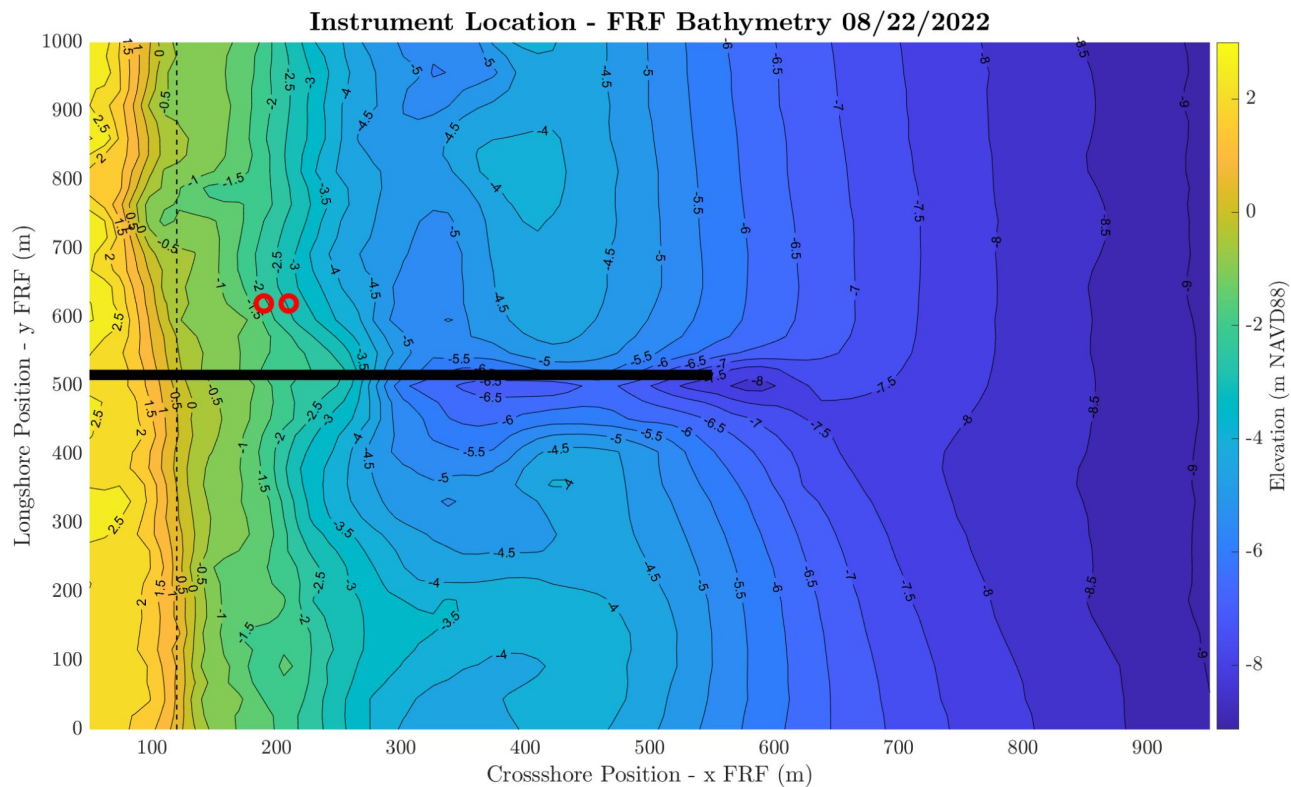


Data Collection Plan

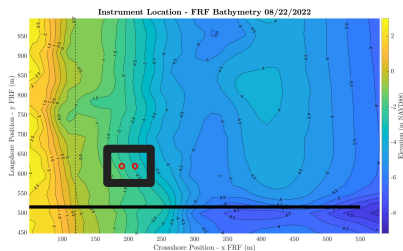
Subsurface
Profiles
And
Multispectral-Drones



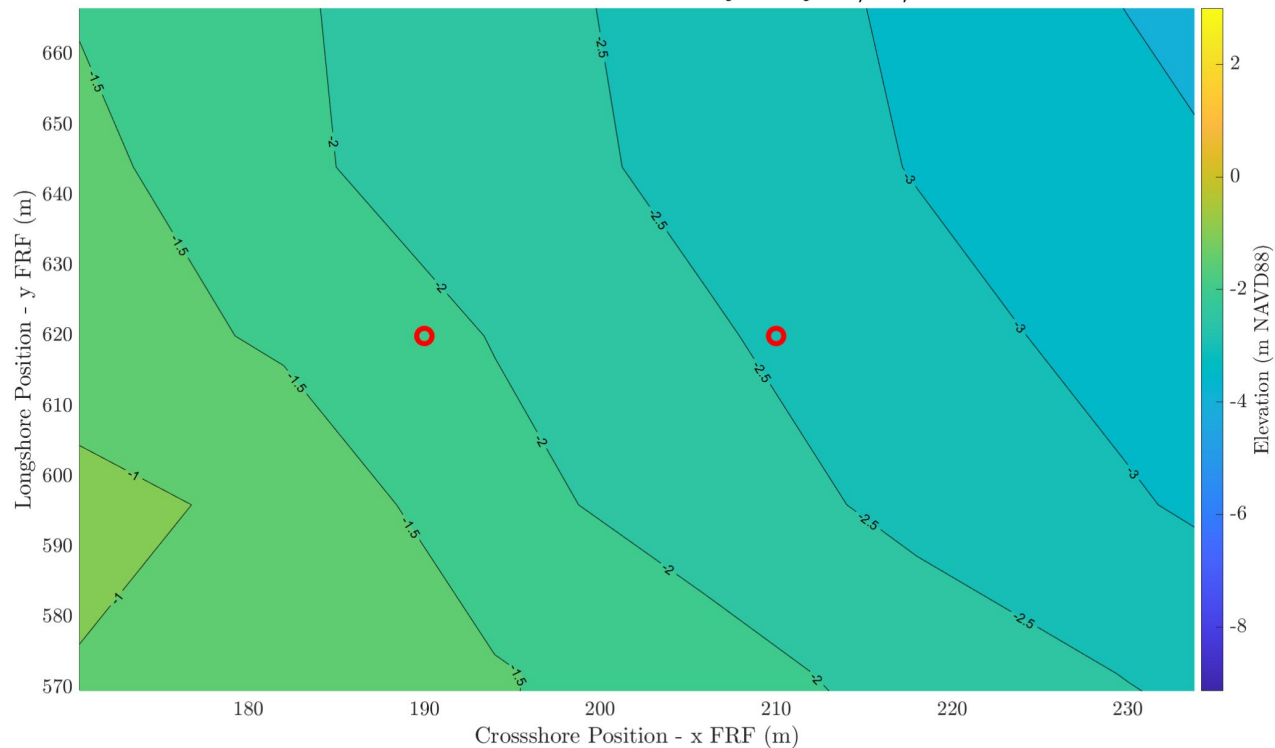
Data Collection



Data Collection



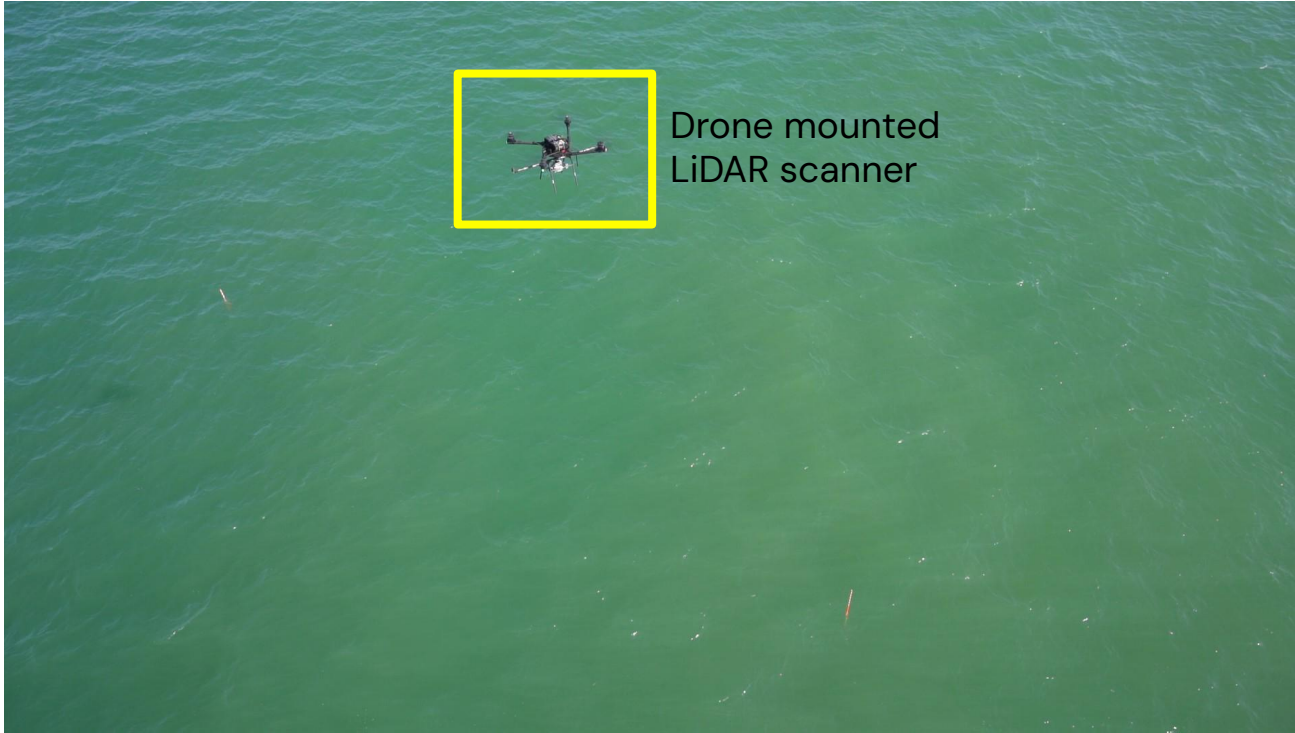
Instrument Location - FRF Bathymetry 08/22/2022



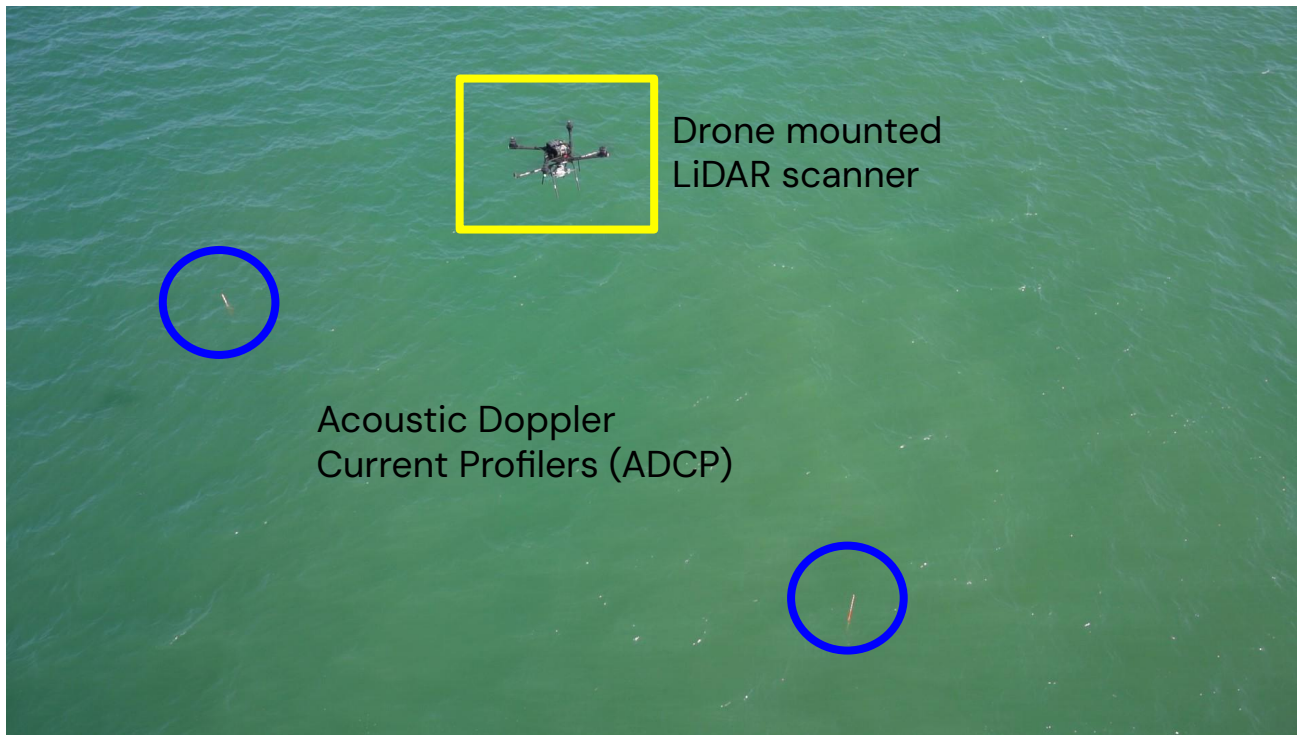
Data Collection



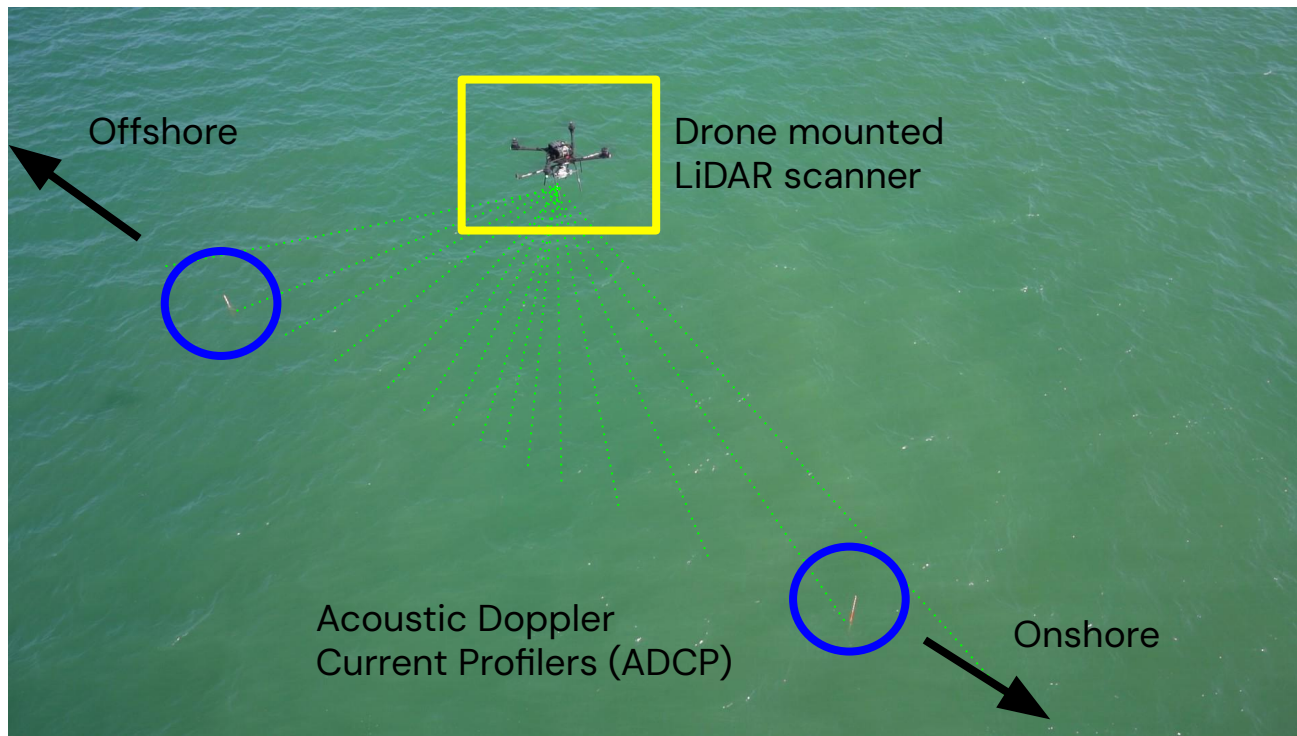
Data Collection



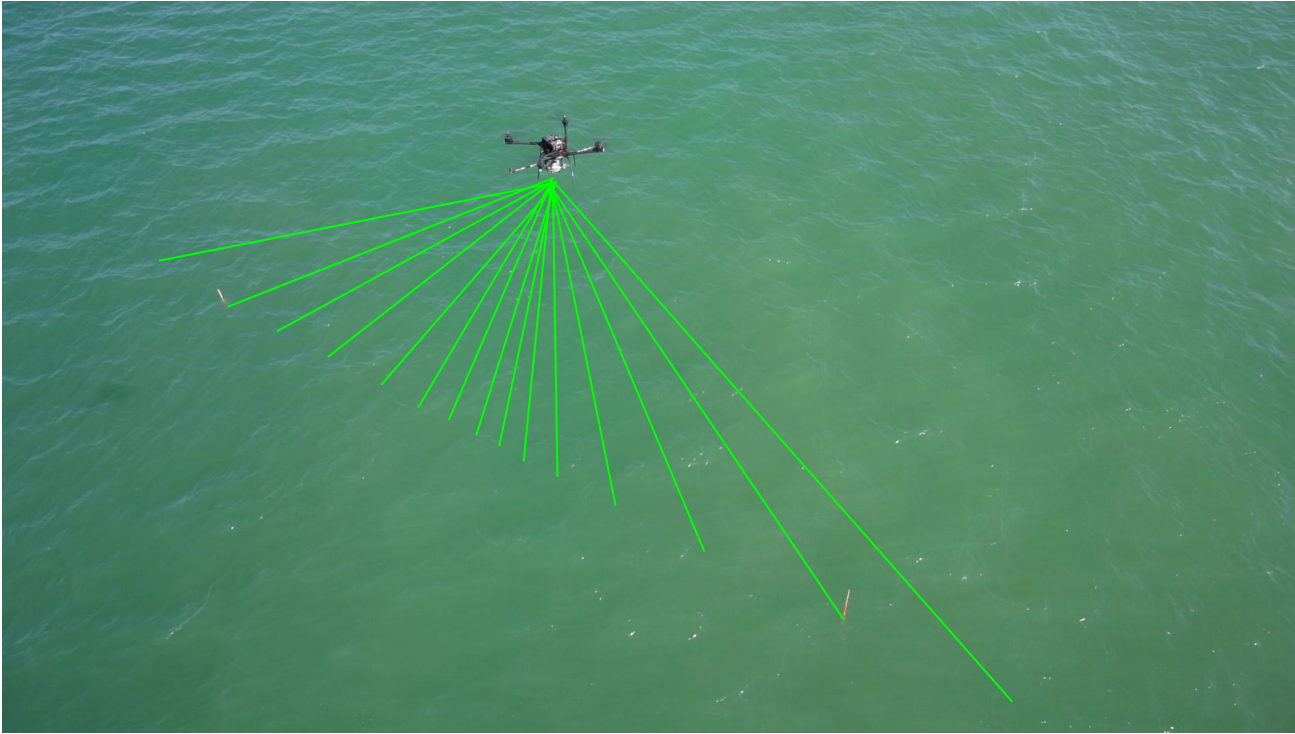
Data Collection



Data Collection



Data Collection



Data Collection- Timeline

September 20

8:00AM

October 20

ADCP Deployed



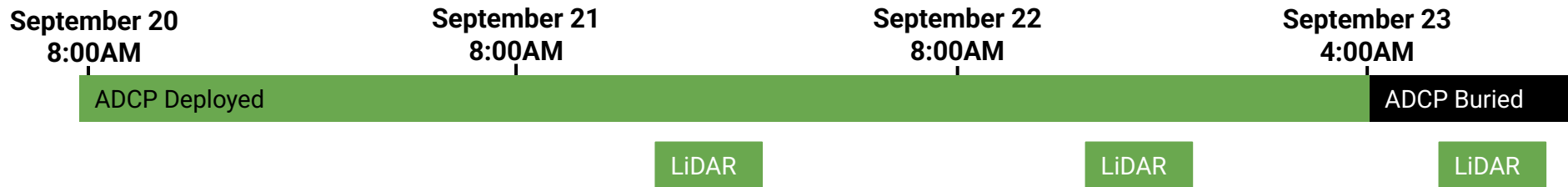
Data Collection– Timeline



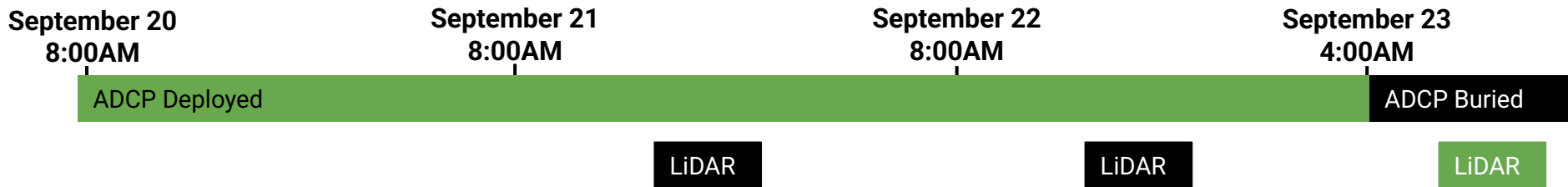
Data Collection- Timeline



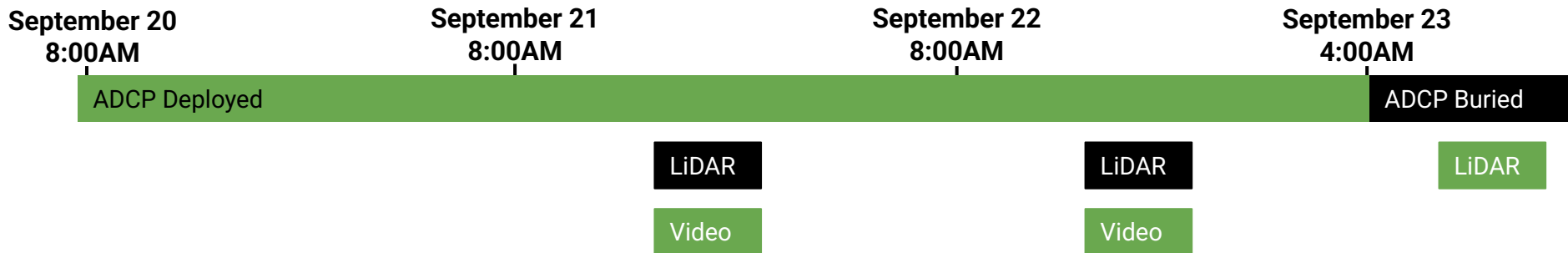
Data Collection– Timeline



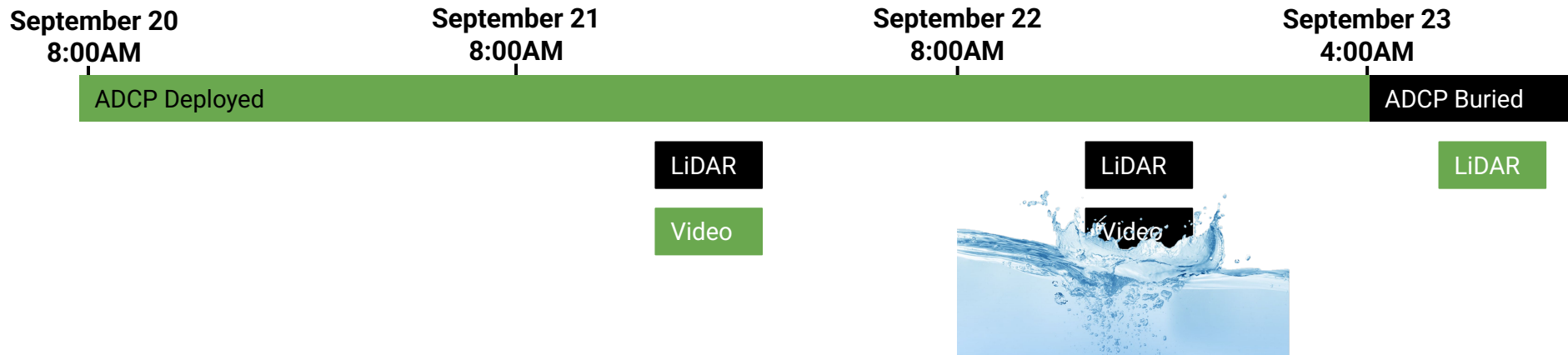
Data Collection– Timeline



Data Collection– Timeline



Data Collection- Timeline



Data Collection - Timeline

September 20
8:00AM

September 23
4:00AM

ADCP Deployed

ADCP Buried

October 20

September 20
8:00AM

September 23
8:00AM

ADCP Deployed

September 23
4:00AM

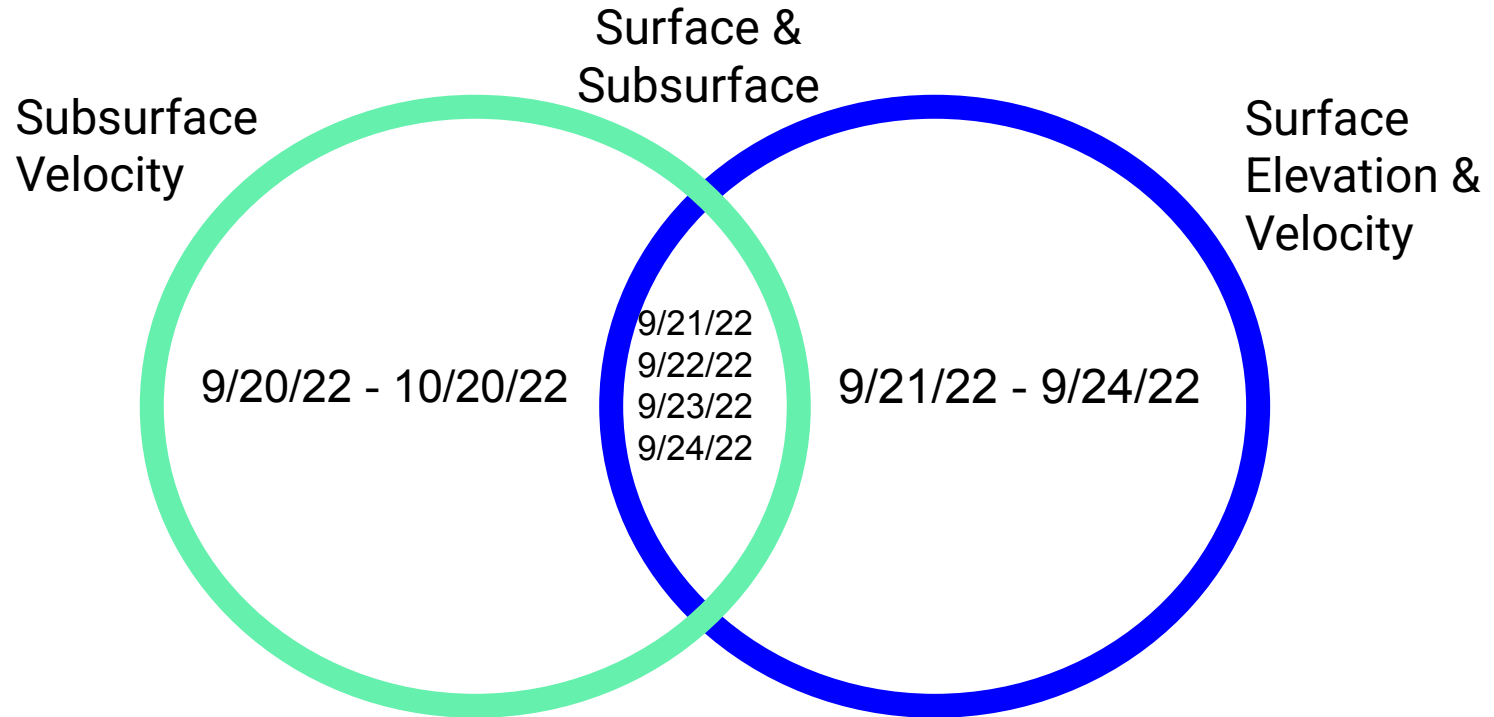
ADCP Buried

LiDAR

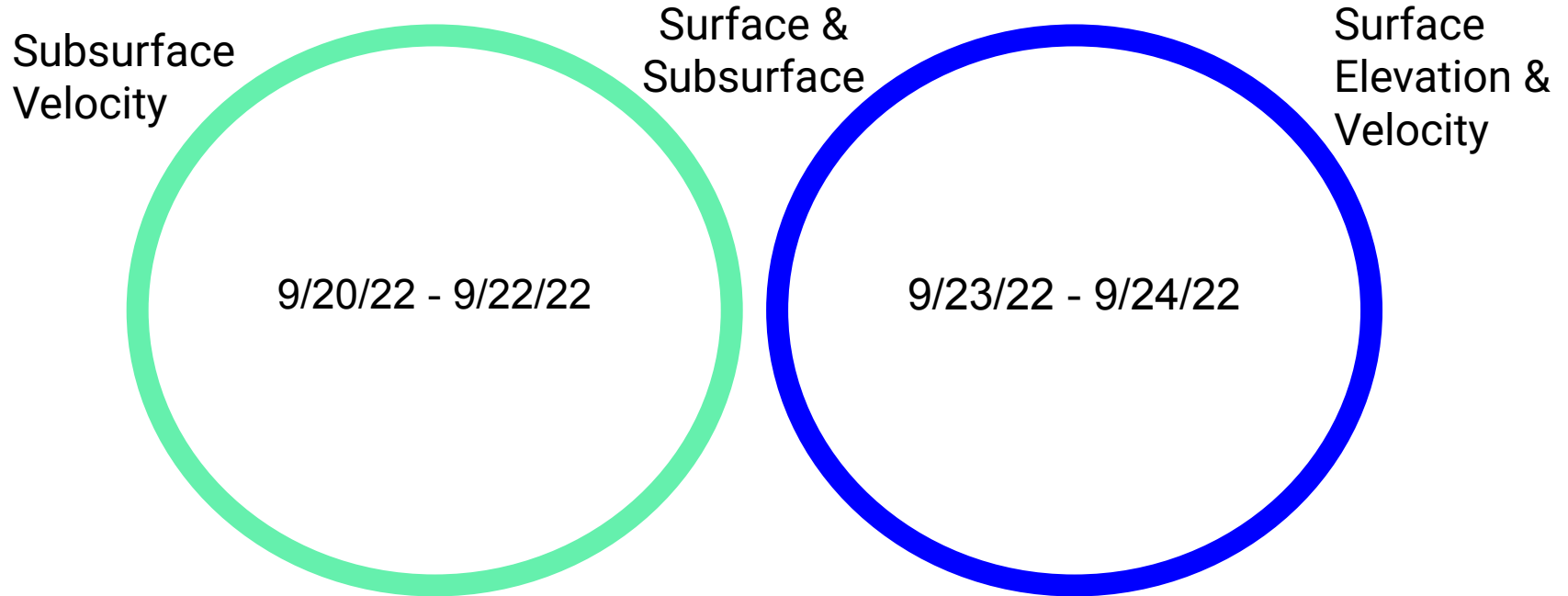
LiDAR

Video

Theoretical LiDAR-ADCP Data Overlap

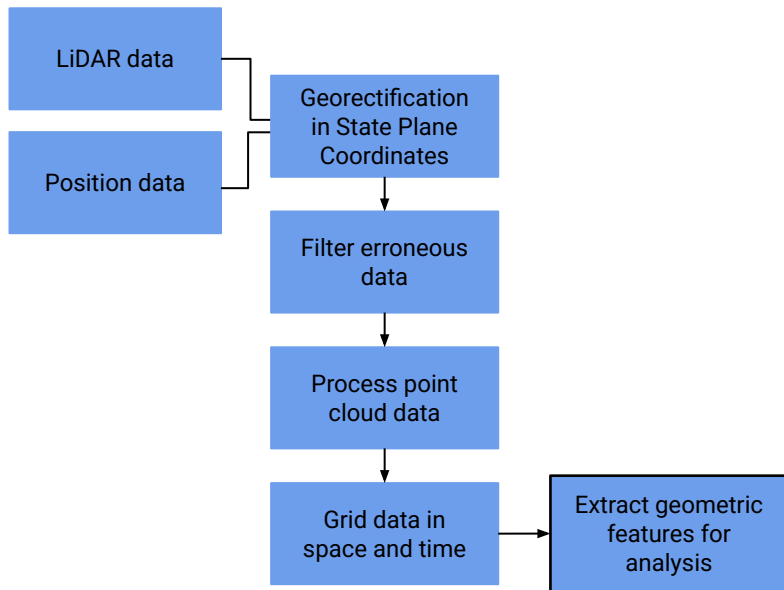


Actual LiDAR-ADCP Data Overlap

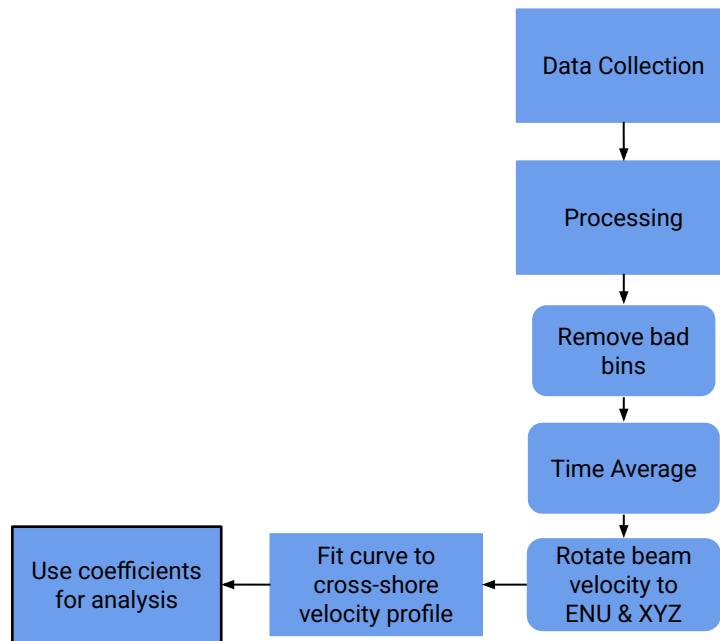


Data Processing- Overview

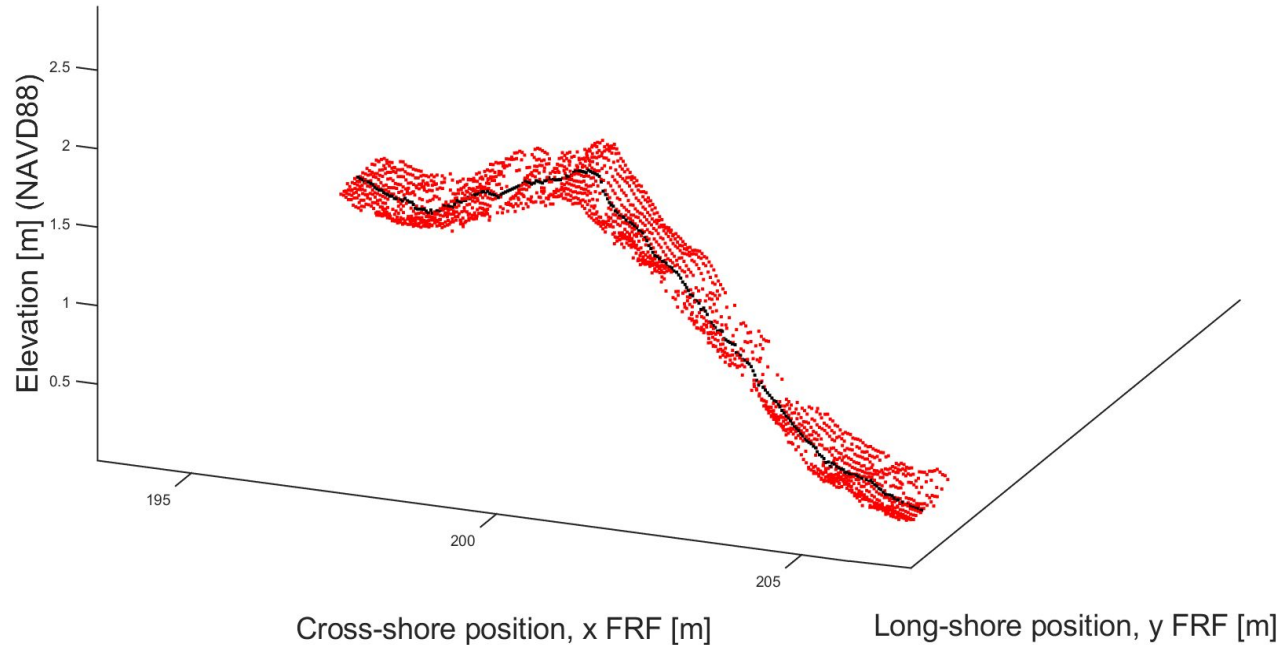
LiDAR



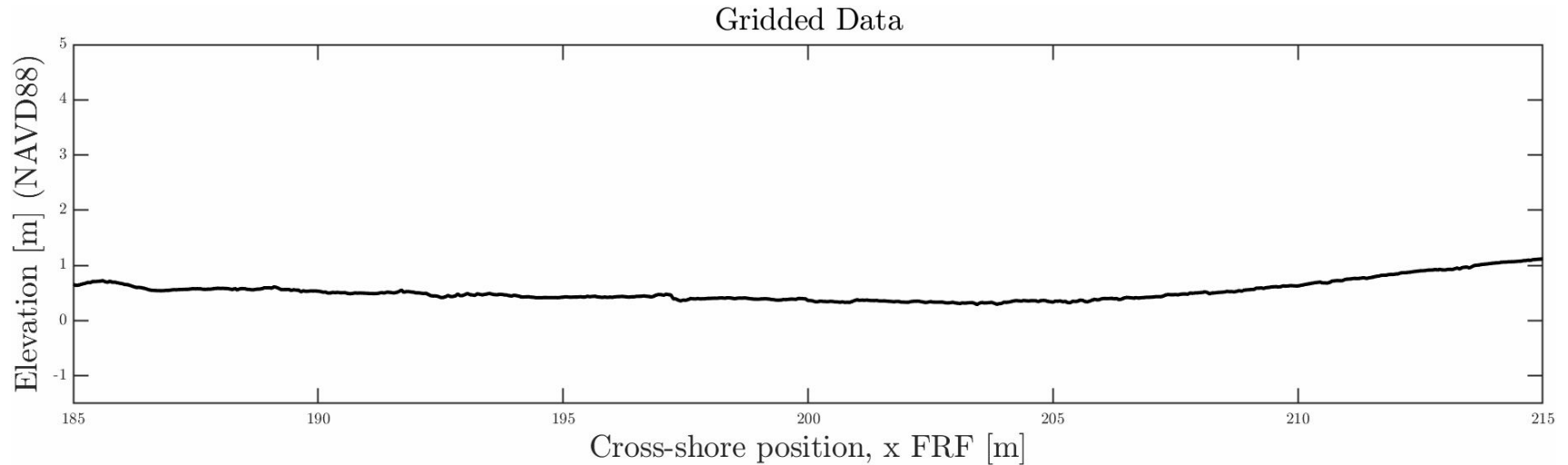
ADCP



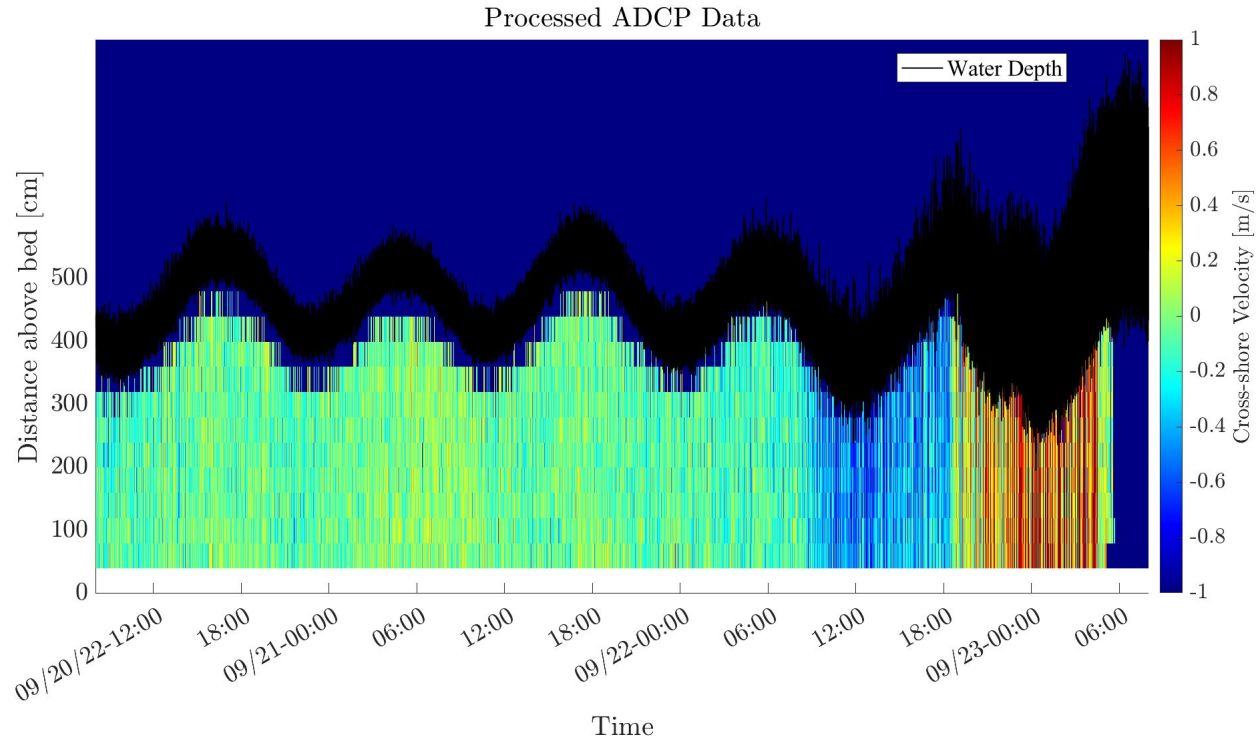
Results– Surface Elevation



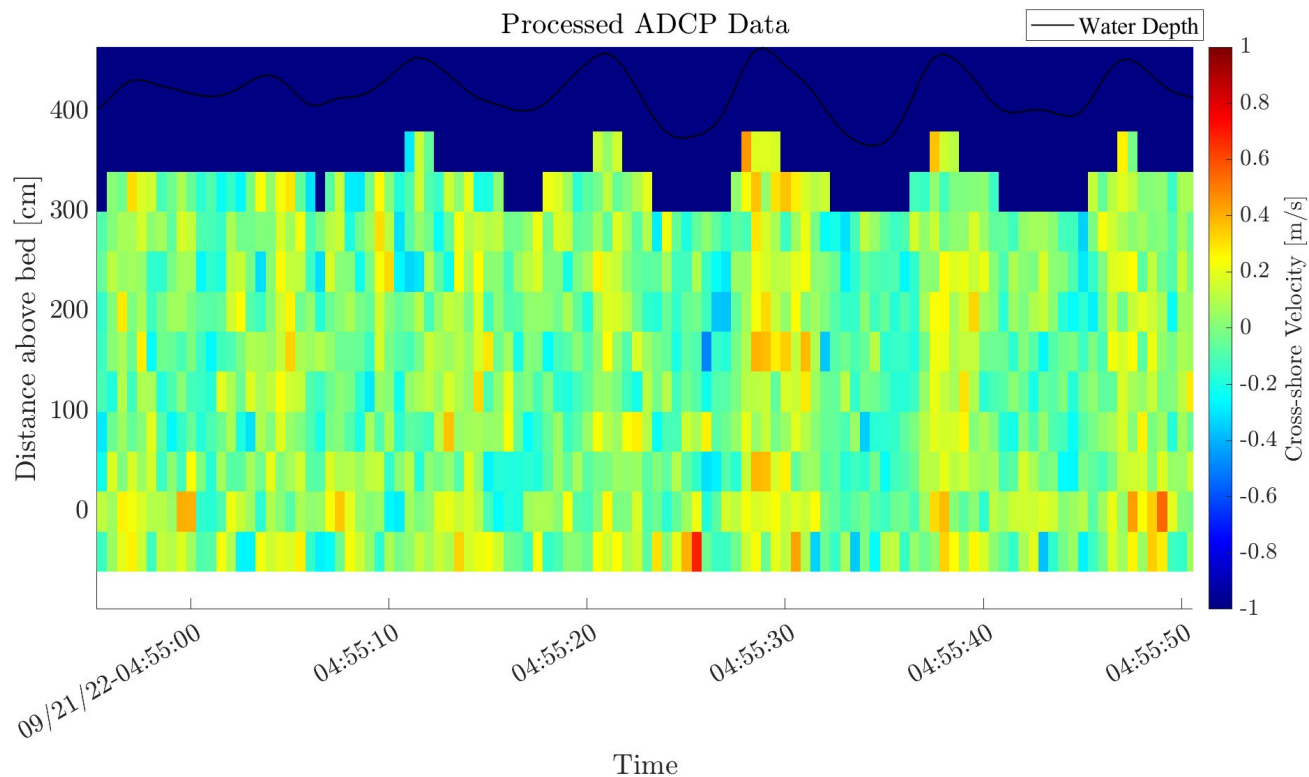
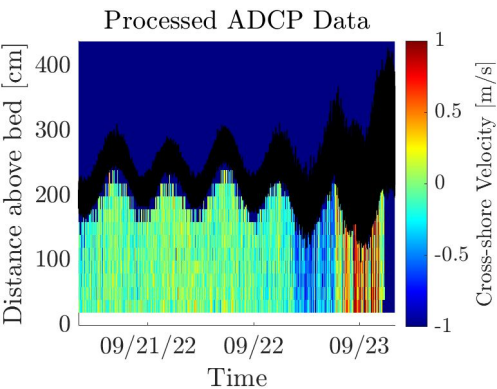
Results– Surface Elevation



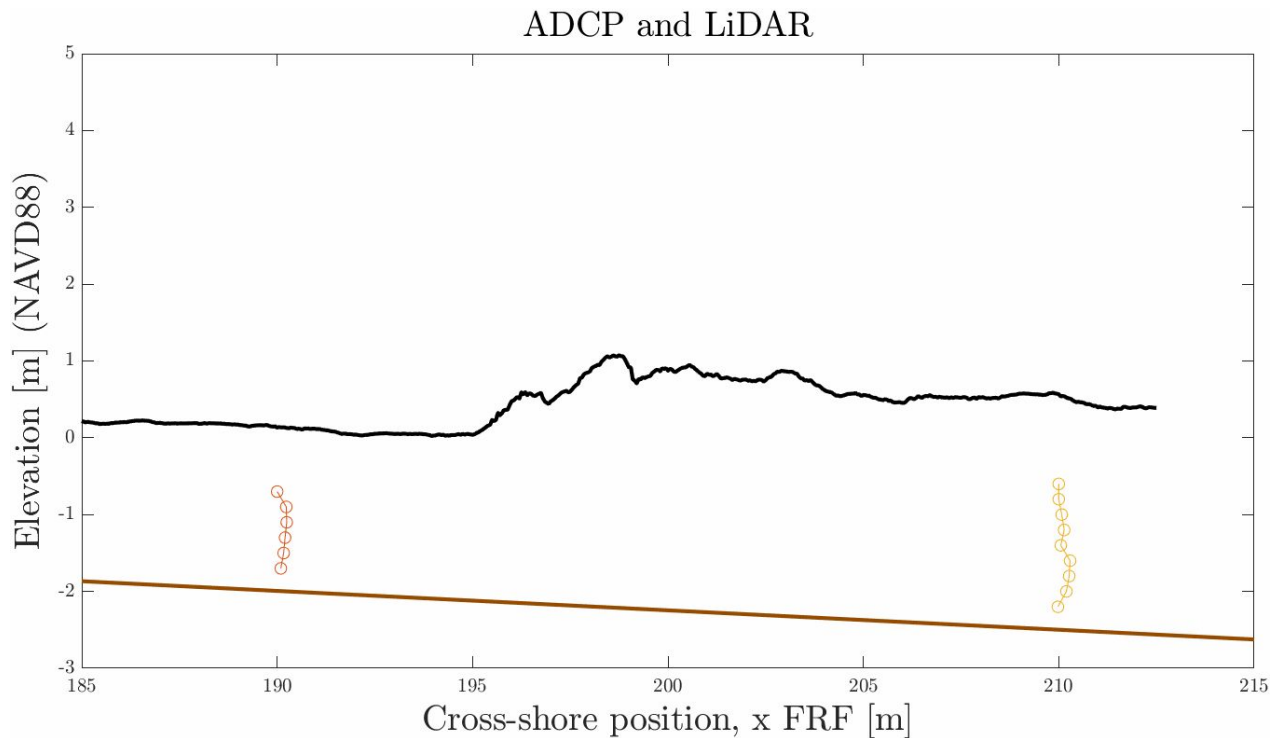
Results– Subsurface Velocity



Results– Subsurface Velocity

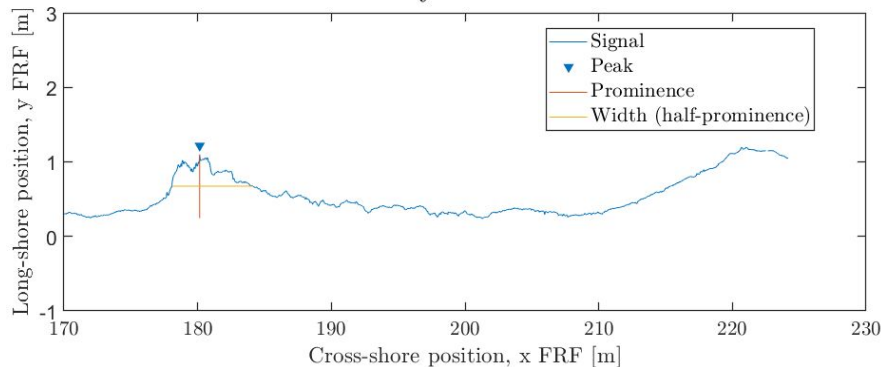


Results– Combined *Not Synchronous*



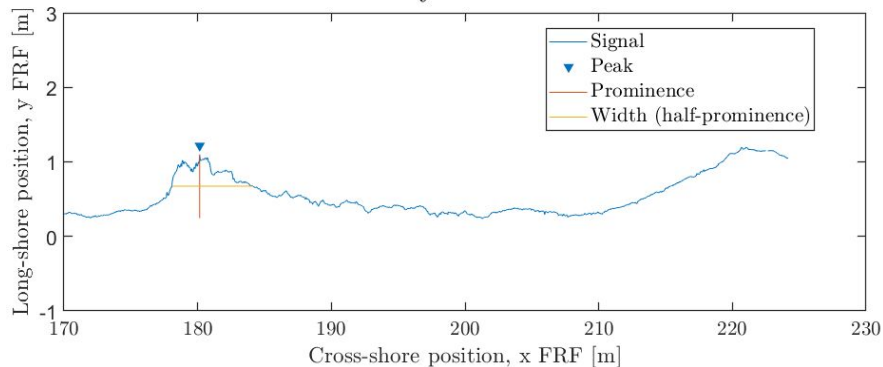
ADCP-LiDAR Analysis | LiDAR Feature Extraction

Wave Geometry Feature Extraction



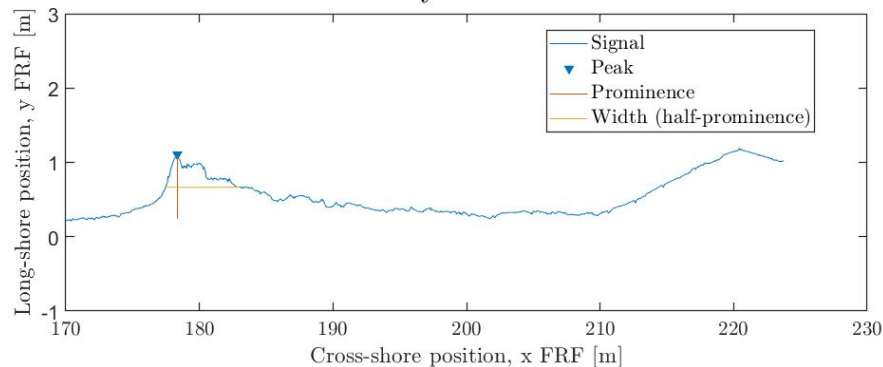
ADCP-LiDAR Analysis | LiDAR Feature Extraction

Wave Geometry Feature Extraction



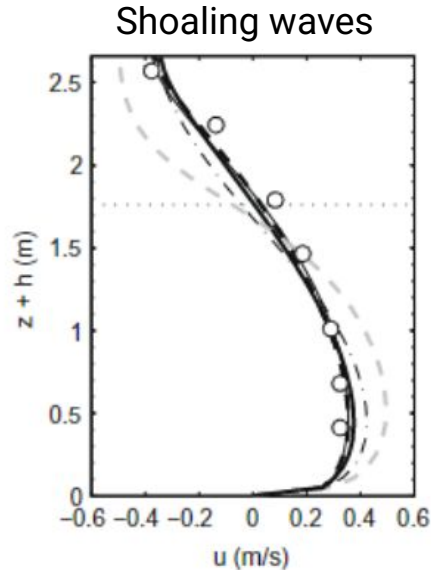
$t=1$

Wave Geometry Feature Extraction

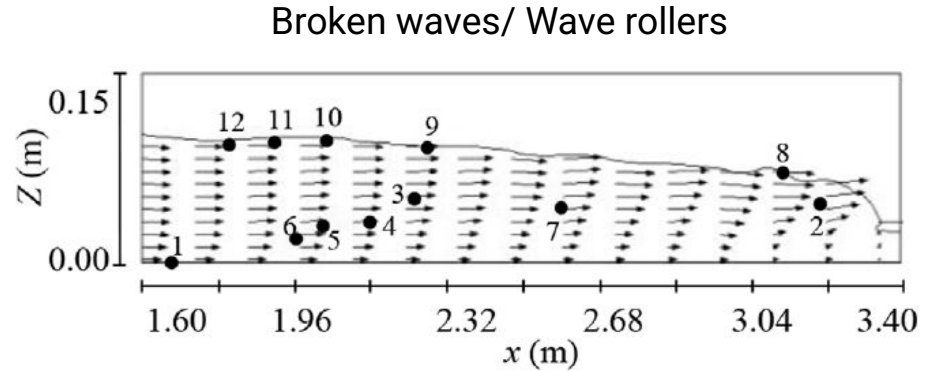


$t=2$

ADCP-LiDAR Analysis | ADCP Curve Fit

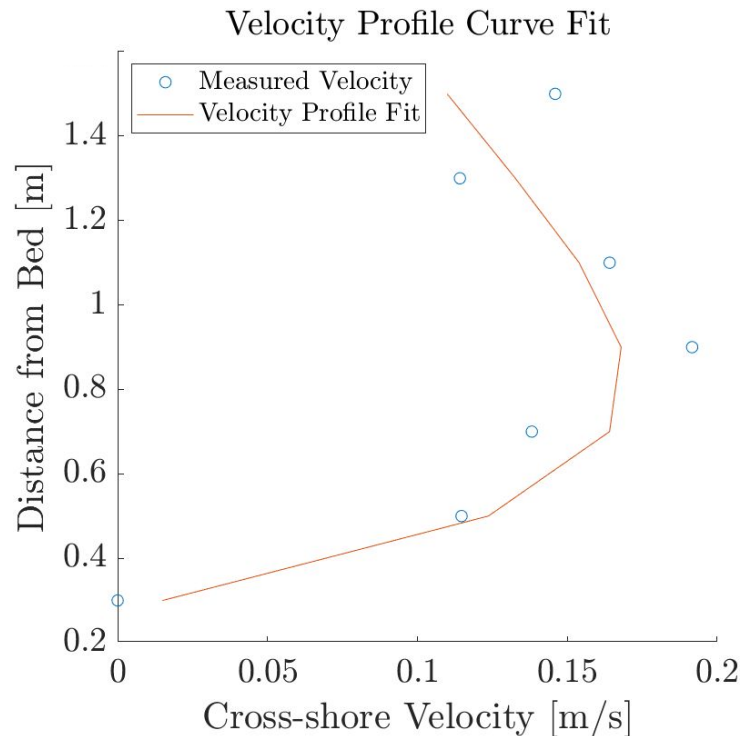


Uchiyama et. al., 2010



Yang 2018

ADCP-LiDAR Analysis | ADCP Curve Fit



$$f(x) = a \cdot \exp(b \cdot x) + c \cdot \exp(d \cdot x)$$

$f(x)$ = Cross-shore velocity

x = Depth above bed

a - d = Coefficients

Coefficients:

$a = 946.1$

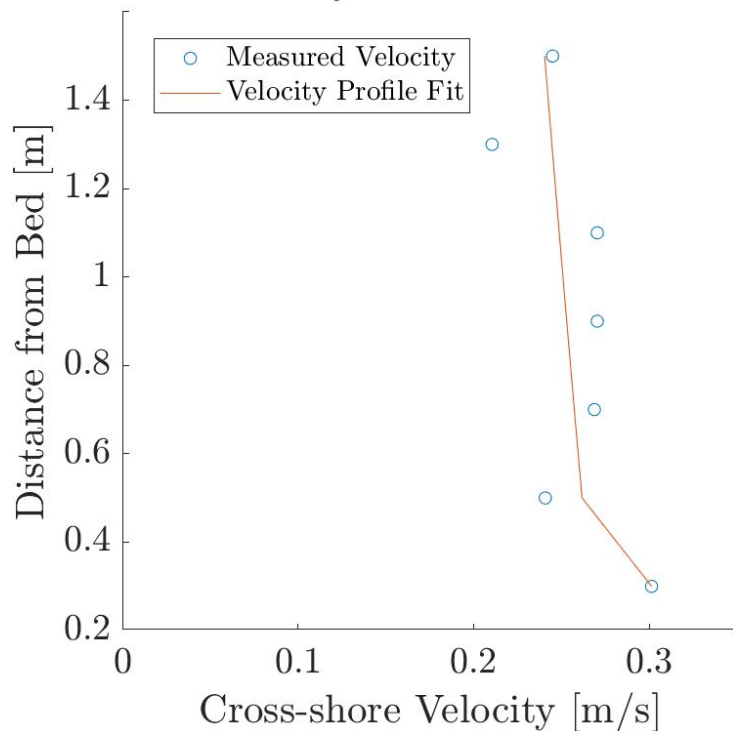
$b = -1.836$

$c = -946.9$

$d = -1.839$

ADCP-LiDAR Analysis | ADCP Curve Fit

Velocity Profile Curve Fit



$$f(x) = a \cdot \exp(b \cdot x) + c \cdot \exp(d \cdot x)$$

$f(x)$ = Cross-shore velocity

x = Depth above bed

a - d = Coefficients

Coefficients:

$$a = 1.315 \times 10^7$$

$$b = -63.49$$

$$c = 0.5462$$

$$d = -0.049$$

Conclusion

- Preliminary results show that surface LiDAR and ADCP measurements can quantify surface-subsurface relationships
- Created a processing pipeline for LiDAR and ADCP data
- Increase the accessibility to subsurface velocities
- Limitations:
 - 2D LiDAR: Waves with incident angle different from scan transect “disappear”
- Improved methods moving forward
 - Integrated LiDAR and RGB camera on one UAV
 - More robust velocity profile fitting function

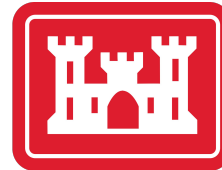
References

Wave–Current Interaction in an Oceanic Circulation Model with a Vortex-Force Formalism: Application to the Surf Zone | Elsevier Enhanced Reader. <https://doi.org/10.1016/j.ocemod.2010.04.002>. Accessed 29 Nov. 2022.

Yang, Shaolin, et al. “Numerical Study on Characteristics of Dam-Break Wave.” *Ocean Engineering*, vol. 159, July 2018, pp. 358–71. *ScienceDirect*, <https://doi.org/10.1016/j.oceaneng.2018.04.011>.

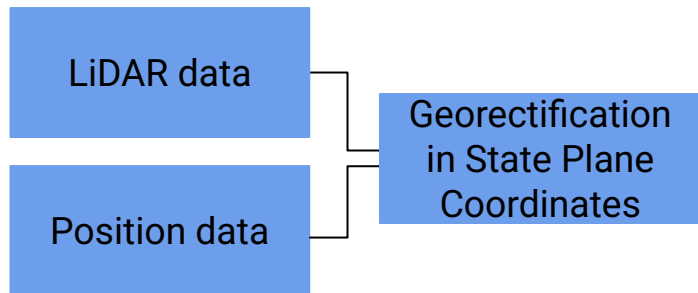
Thank you!

Questions?

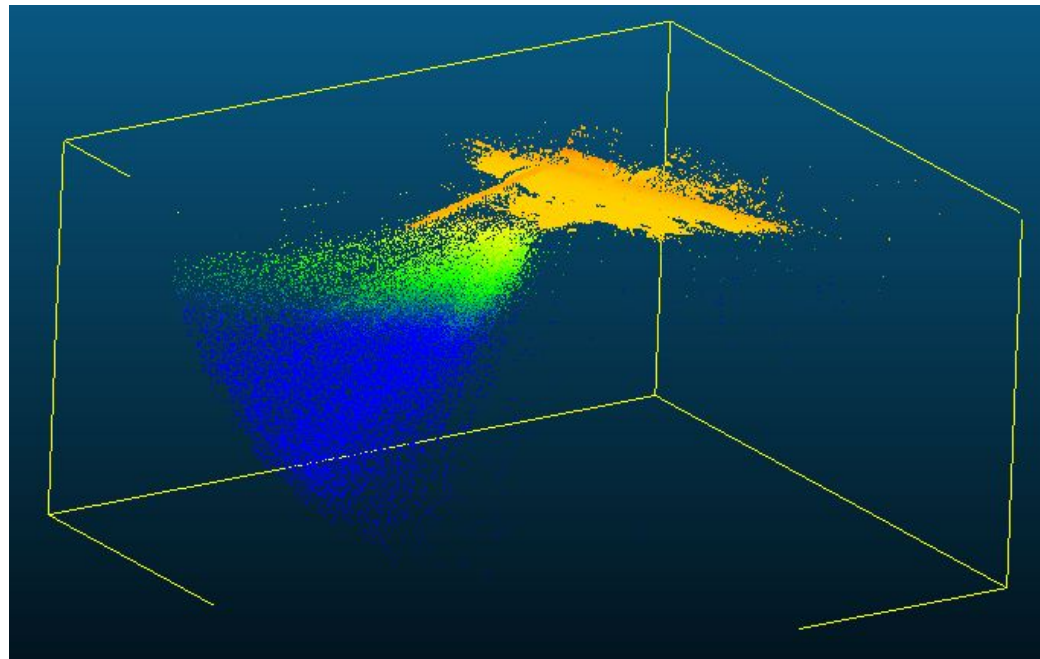
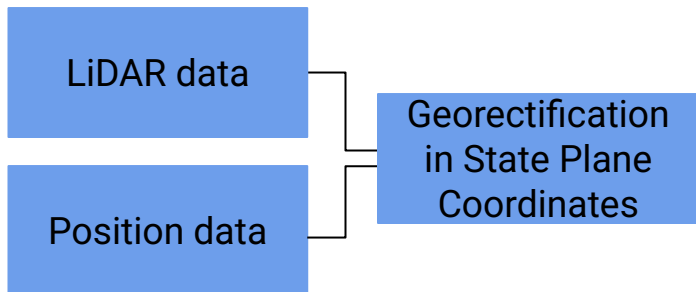


Extra Slides

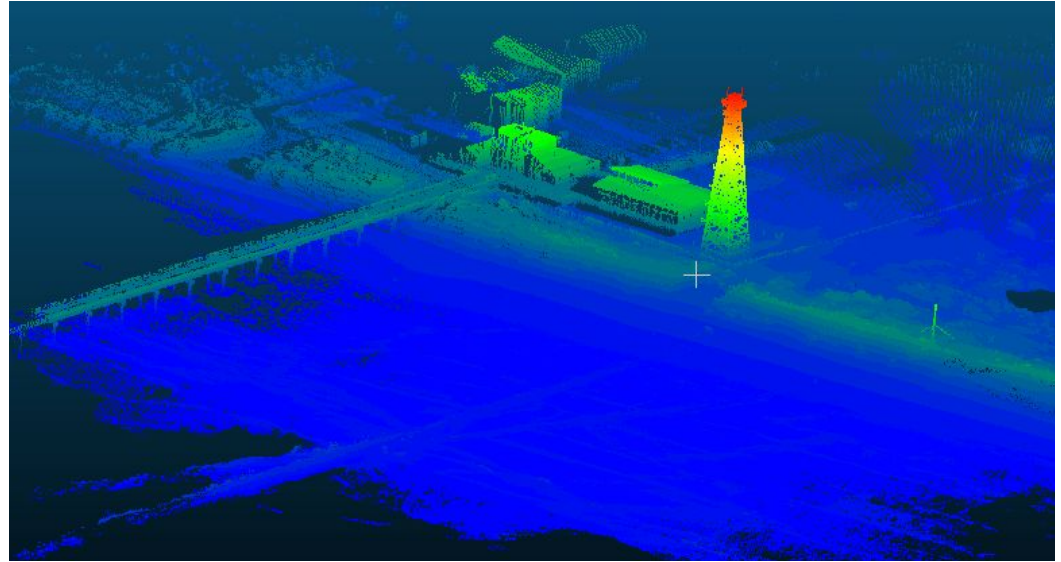
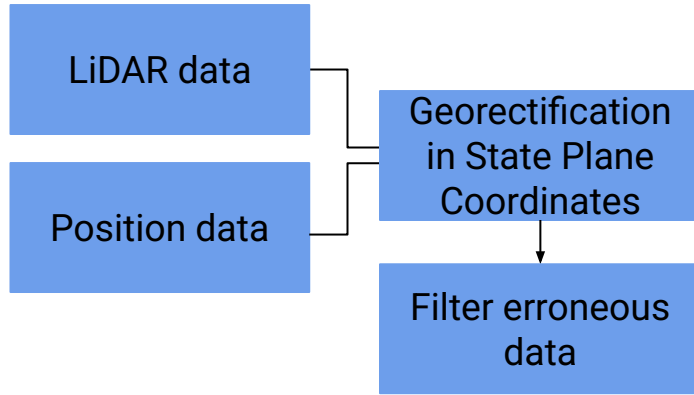
Data Processing- LiDAR



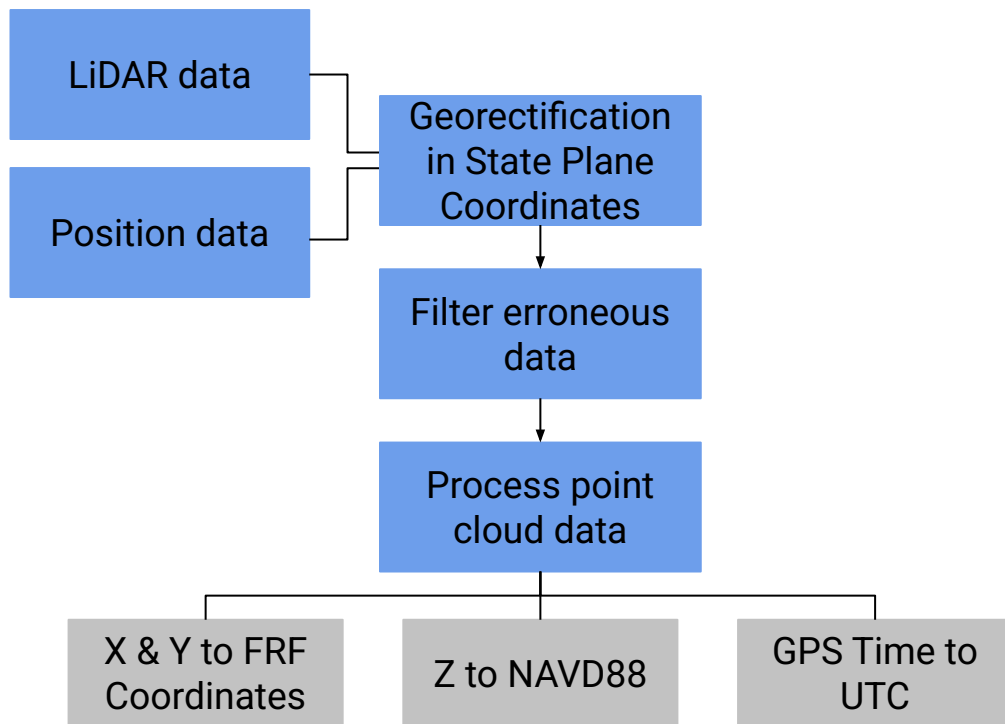
Data Processing- LiDAR



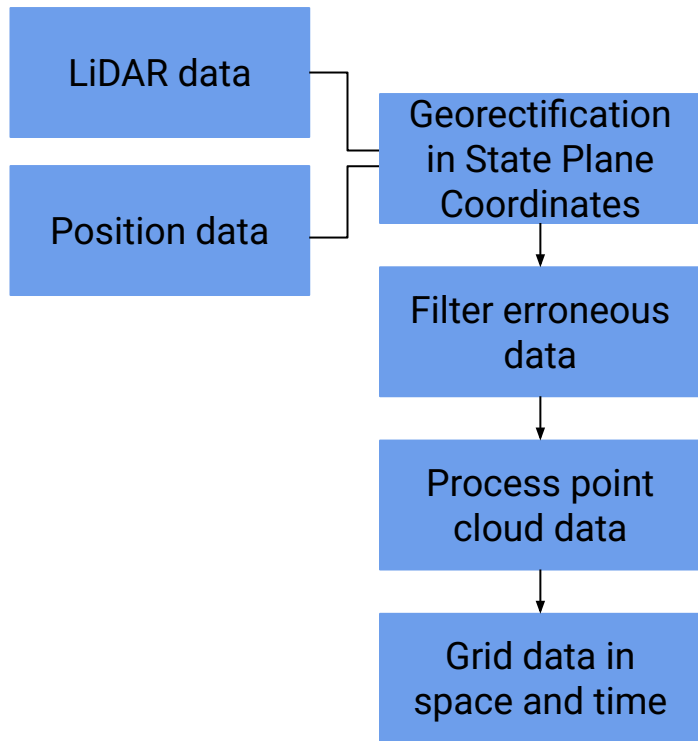
Data Processing- LiDAR



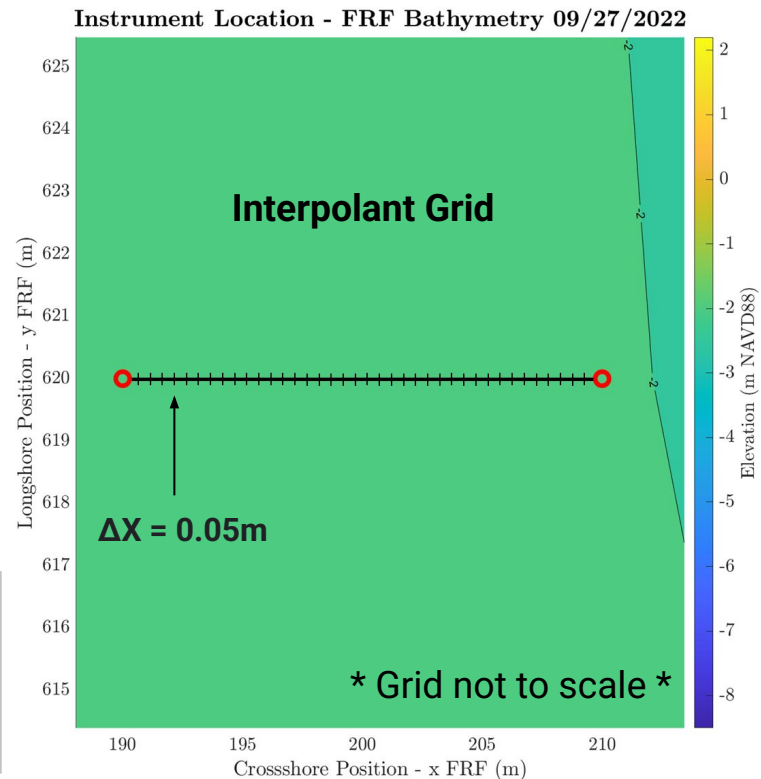
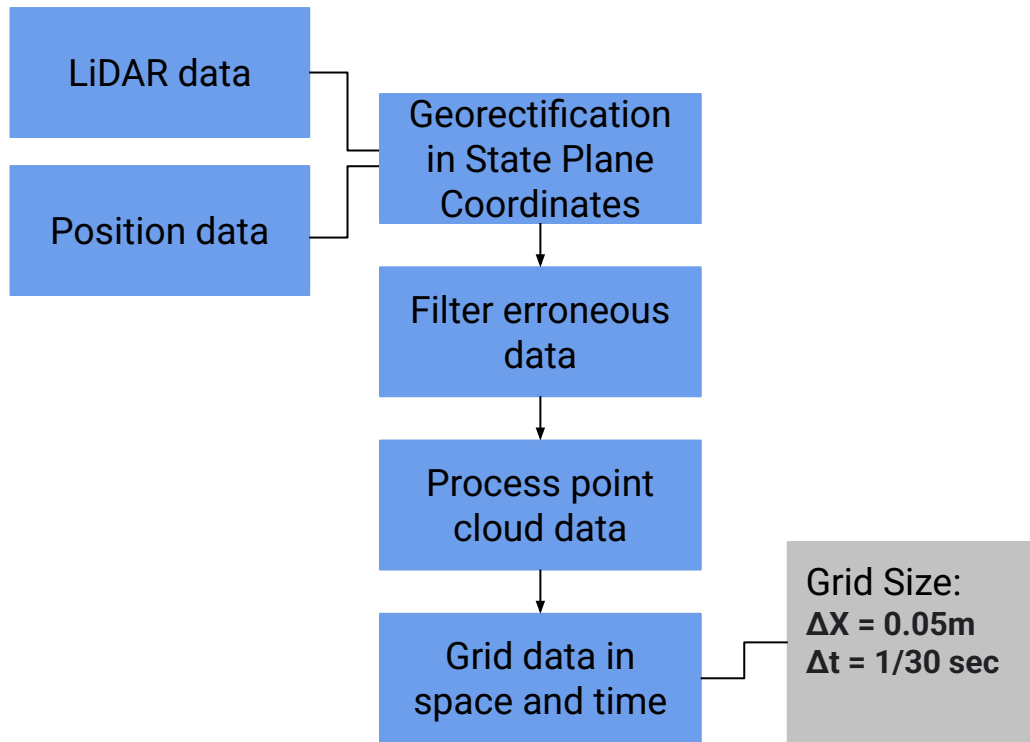
Data Processing- LiDAR



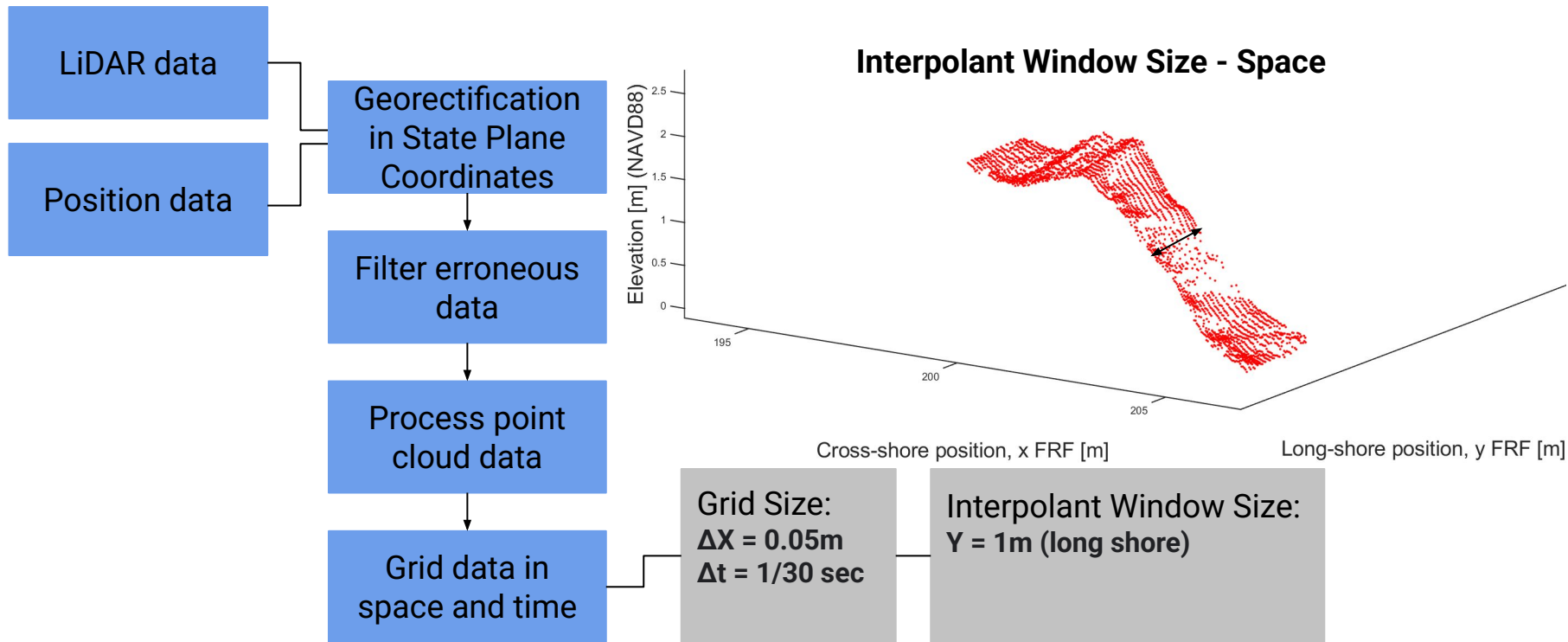
Data Processing- LiDAR



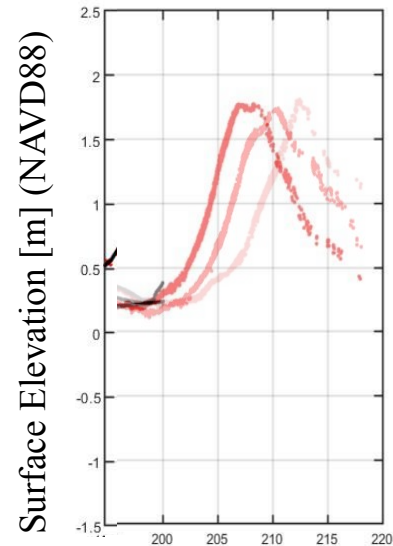
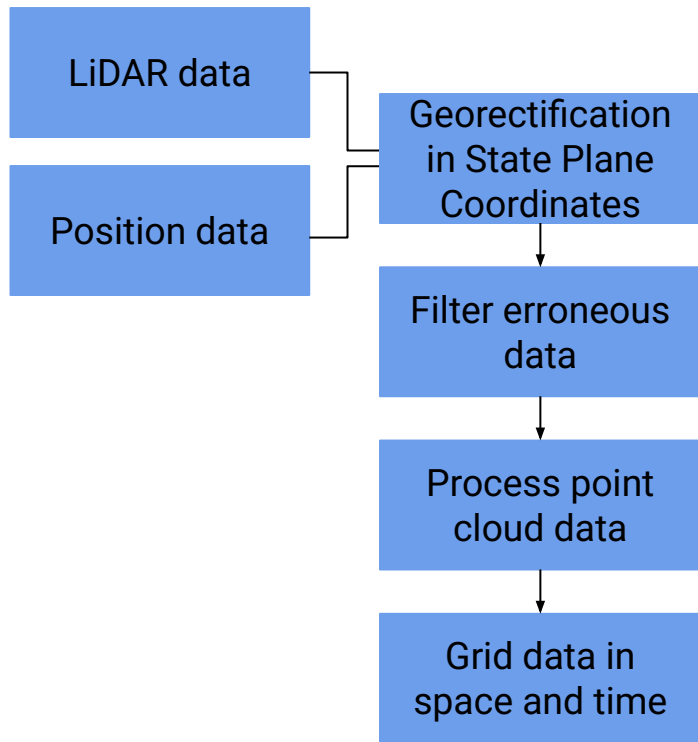
Data Processing- LiDAR



Data Processing- LiDAR



Data Processing- LiDAR

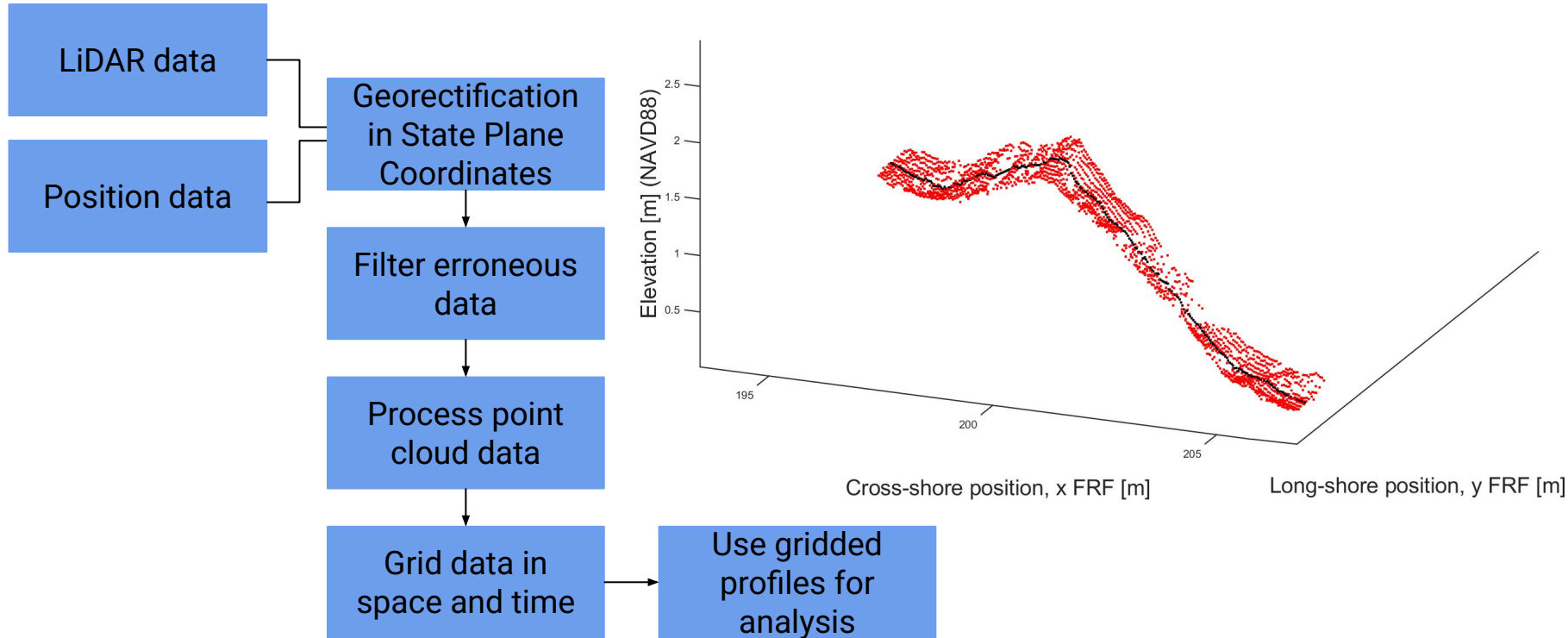


FRF X Coordinates (Cross-shore) [m]

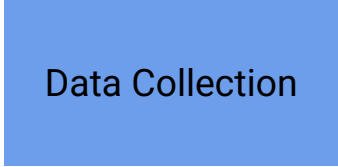
Grid Size:
 $\Delta X = 0.05\text{m}$
 $\Delta t = 1/30 \text{ sec}$

Interpolant Window Size:
 $Y = 1\text{m (long shore)}$
 $T = 2.5 / f_s \approx 0.083 \text{ sec}$
(2 frames forward, 2 frames back)

Data Processing- LiDAR

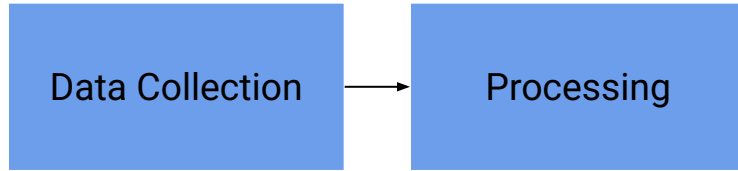


Data Processing- ADCP

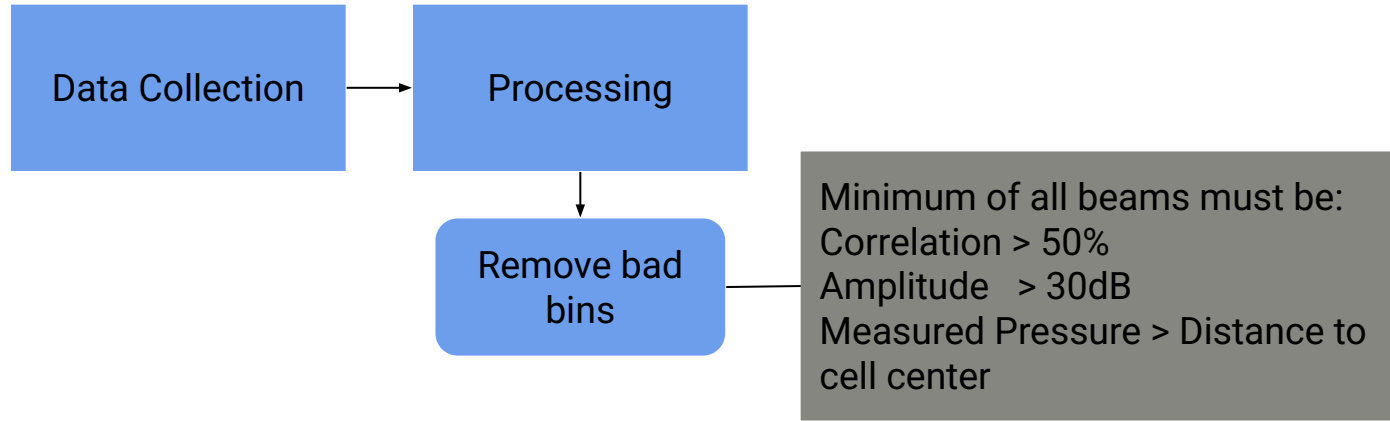


Data Collection

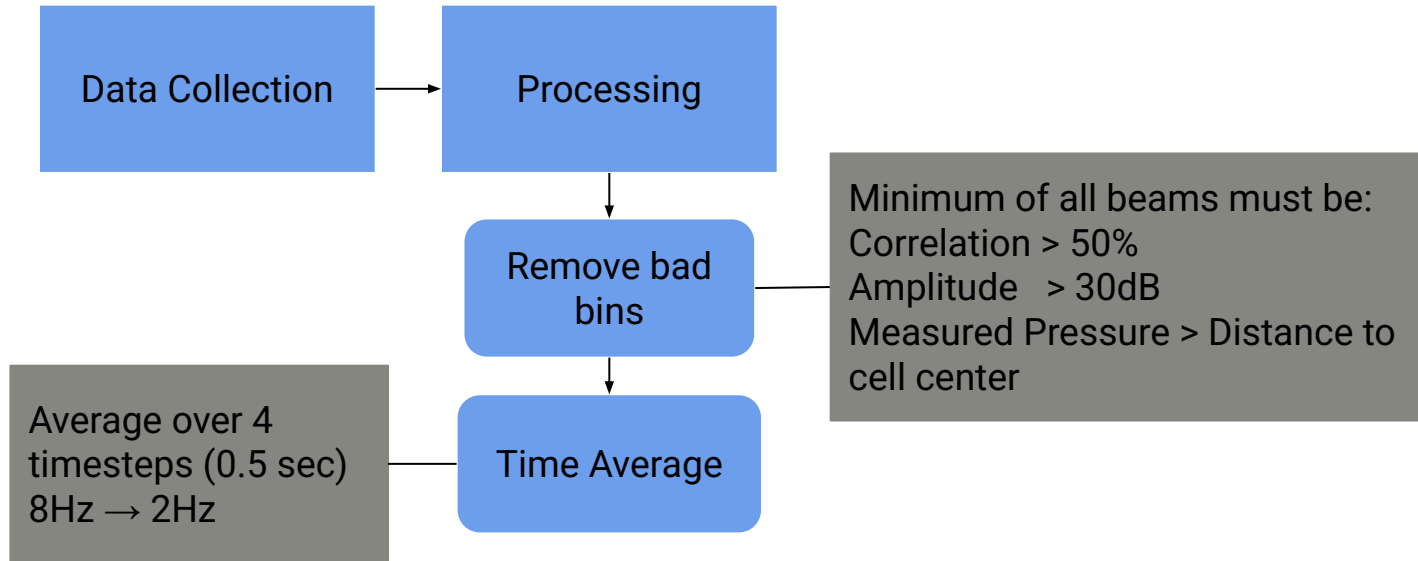
Data Processing- ADCP



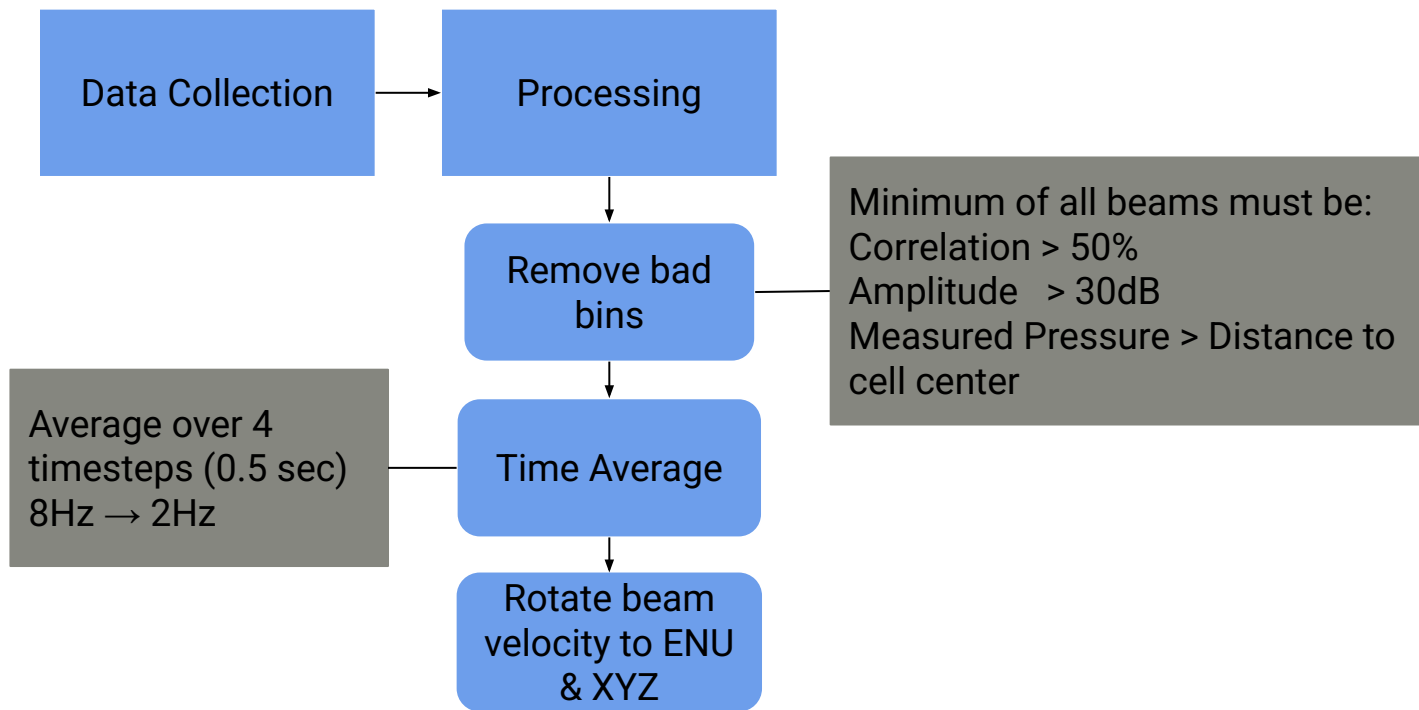
Data Processing- ADCP



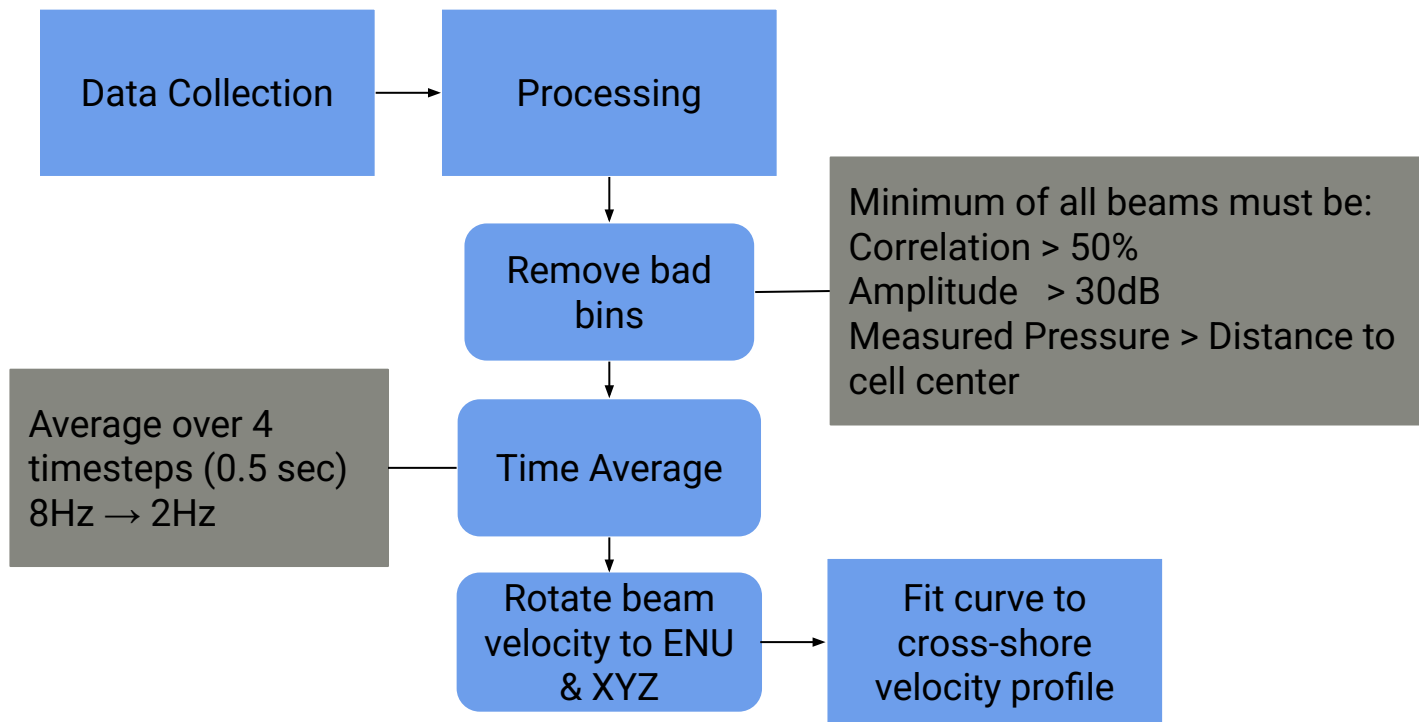
Data Processing- ADCP



Data Processing- ADCP



Data Processing- ADCP



Data Processing- ADCP

